

Analysis of satellite streaks in astronomical images down to low signal-to-noise ratios with deep learning

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ABSTRACT

The drastic increase in the number of satellites in low-earth orbit (LEO) poses unprecedented challenges for time-domain optical and near-infrared astronomy. We present a deep learning pipeline to **analyse** satellite streaks in short-exposure, single-band images, based on targeted observations of LEO satellites with the global network of Unistellar telescopes. **We develop a simulation software that injects realistic streaks onto telescope images, and we train (1) traditional convolutional neural networks (CNN) to classify images including streak candidates among large data sets, (2) fully-convolutional networks (UNet) to conduct the pixel-level reconstruction of satellite streaks. Besides reducing the inference time on large, unbalanced data sets, this approach helps quantify the performance as a function of streak properties, and allows us to optimize the mask reconstructions down to the lowest signal-to-noise ratios.** The CNN classifier reaches a true-positive rate (TPR) of nearly 90% on real satellite streaks while keeping the false-positive rate (FPR) below 1%. The UNet also reaches excellent performance for the segmentation of images selected by the CNN. For streaks with a peak S/N above 5, we recover >80% pixels within the ground-truth satellites masks, for a contamination level of maximum 10 false positive pixels per image (FPR per pixel of 0.001%). For 90% of streak pixels recovered by the UNet, the precision of output satellite masks is >96% at $(S/N)_{\text{peak}} \in [5;8]$, and nearly 100% at $(S/N)_{\text{peak}} \geq 8$. We develop a separate UNet to optimize the reconstruction of the faintest streaks **partially below the background noise levels**, including optimization of the architecture, and tests of various pre-processing methods **such as image denoising**. With this network, the precision at 90% recall increases to 40.7% at $(S/N)_{\text{peak}} < 5$, and we obtain accurate predictions on the streak endpoint coordinates. Moreover, the UNet performs well on deep images from the Subaru 8-m class telescope without any fine-tuning. **Overall, the pipeline allows (1) accurate measurements of satellite positions over broad (S/N) ranges, and (2) efficient generation of realistic data sets for neural network retraining, which makes it promising to mitigate the impact of satellite constellations on wide-field astronomical images from different telescopes.**

Key words. Surveys – Techniques: image processing – Methods: numerical

1. Introduction

The number of artificial satellites in Low-Earth Orbit (LEO) for wireless communications has been growing exponentially since the last few years and is expected to reach a total number of a few 10 000 in 2030. Given their substantial reflection of Sun light, LEO satellites leave bright, elongated signatures (referred to as "streaks" hereafter) in images of the sky, particularly at low elevation and close to twilight (McDowell 2020). These streaks are creating increasingly complex problems for the image processing pipelines and are raising major concerns for the future of ground-based astronomical research (e.g., IAU press release¹). Hainaut & Williams (2020) studied the impact of satellite constellations on astronomical imaging and spectroscopy in the optical and infrared domains, finding that deep, wide-scale imaging surveys on the largest telescopes would be most significantly affected. In particular, a total of 30–40% of images from the forthcoming *Rubin* Observatory Legacy Survey of Space and Time (LSST) would be impacted at the beginning and end of the night, with saturation of the detectors. Tyson et al. (2020) further quantified the maximal satellite brightnesses that would allow to mitigate electronic ghosts on the *Rubin* Observatory camera sensors, and to correct the nonlinear image artifacts to a sufficient level.

Flagging and masking out satellite streaks in astronomical images is particularly challenging for time-domain surveys such as the *Rubin* LSST. In these cases, since detection alerts are obtained from single-exposure images, the identification and masking of streaks can not be conducted through temporal filtering, and needs to rely purely on the streak morphological signatures. Recent softwares such as LEOSatpy² or SatMetrics³ have been developed to identify pixels affected by such streaks. Assuming nearly perfect linearity, these softwares usually rely on the Hough Transformation method (Hough 1962) to extract the streak segments (e.g., Xu et al. 2015). Extensions such as the Kernel-based Hough Transforms (e.g., Fernandes & Oliveira 2008) allow to reduce the number of contaminant line detections, and proved successful to accurately mask bright streaks detected in Hyper Suprime-Cam (HSC) images⁴.

Over the last decade, deep learning pipelines have proven highly-efficient to select and analyze extended objects in astronomical images. Deep learning pipelines have, for instance, proven to enhance the extraction of cosmic ray traces in individual telescope images (e.g., Zhang & Bloom 2020; Xu et al. 2023) compared to state-of-the-art edge-detection techniques (e.g., LA Cosmic, van Dokkum 2001), thanks to higher flexibility to di-

² <https://github.com/CLEOsat-group/leosatpy>

³ <https://github.com/uwescience/satmetrics>

⁴ <https://dmtn-197.lsst.io/>

¹ <https://www.iau.org/news/announcements/detail/ann19035/>

verse cosmic ray shapes. Besides cosmic rays, asteroids and other fast moving objects have characteristic morphological signatures in wide-field images. Duev et al. (2019) trained an ensemble of convolutional neural networks (CNN, LeCun & Bengio 1998) to classify short and long asteroid trails in the individual, 30 sec exposures from the Zwicky Transient Facility (ZTF) time-domain survey. Its application to the ZTF alert stream enables efficient selection of candidate near-Earth asteroids, with limited need for human visual inspection. Later on, this work was extended to the detection of fainter trails in ZTF exposures (Wang et al. 2022), and to selection of near-Earth objects in images from the Asteroid Terrestrial-impact Last Alert System (ATLAS) telescopes (Chyba Rabeendran & Denneau 2021). Similar efforts have been pursued for space-based imaging surveys. Lieu et al. (2019) developed a binary classifier pre-trained on the ImageNet data set and fine-tuned on simulations to select asteroids in Euclid-like images. Later on, Kruk et al. (2022) took a step forward and searched for asteroids trails in high-resolution images from the *Hubble* Space Telescope (HST), by training the Google Cloud AutoML Vision algorithm (Zoph & Le 2017) on HST images labelled by citizen scientists. AutoML not only performs classification, but also returns bounding boxes to help localize the trails, and its inference proved stable for the heterogeneous depth of HST single-exposure and stacked images. These studies pave the way to novel supervised machine learning approaches for the characterization of satellite streaks, which have comparable shapes and characteristic lengths.

Recently, Paillassa et al. (2020) developed a fully-convolutional network *MaxiMask* to identify different types of contaminants through semantic segmentation (e.g., Long et al. 2015), in order to minimize the number of spurious detections in deep, single-band images. Contaminants were divided into subclasses ranging from electronic, optical, and post-processing contaminants, to cosmic rays, satellite streaks, and nebulosities. While the versatile pipeline from Paillassa et al. (2020), can in principle focus on flagging pixels belonging to satellite streaks, it was trained on deep, wide-field images containing simultaneously all subclasses of contaminants. Simulating highly-realistic satellite streaks and training separate neural networks for binary segmentation could therefore improve the robustness of the streak reconstructions.

Besides the substantial decrease in inference time, the characterization of satellite streaks with dedicated neural networks opens interesting perspectives to improve model stability with respect, for instance, to observing conditions, to (S/N) regimes, to blending with bright astronomical sources, and to complex effects such as variable surface brightness along the streaks, or discontinuities due to pointing errors. **Recently, Stoppa et al. (2024) combined a UNet with probabilistic Hough transform to conduct satellite streak detection on images from the MeerLICHT telescope, while also removing discontinuities in the output masks. They identify real, detected LEO/GEO streaks that are manually labelled in order to train their UNet. They obtain excellent performance on their test set of MeerLICHT images, with 97% positive pixels correctly predicted, and with robust segmentation in the presence of diffraction spikes. Interestingly, applying their UNet to 200 000 new images results in the detection of >1700 uncatalogued objects. Moreover, Liu et al. (2025) propose a complementary deep learning pipeline addressing streak identification with the ShuffleNet V2 model. While this study focuses on the classification of images with and without streak, and does not explore mask reconstructions, it demonstrates its ef-**

iciency compared to non-machine learning algorithms (e.g. SExtractor).

In this work, we develop a novel supervised deep learning pipeline to analyze astronomical affected by extended streaks from LEO satellites. **To complement previous studies that were either focusing on image classification (e.g., Liu et al. 2025), or did not discuss the evolution of performance as a function of (S/N) (Stoppa et al. 2024), we search for the ingredients to achieve accurate streak reconstruction over broad ranges of ground-truth streak properties. In particular, we optimize performance in the very faint regime, where streaks are partially below the background noise levels, to better quantify and reduce the impact of satellite constellations on wide-field astronomical images (see also Bassa et al. 2022).** Broader applications range from monitoring the efficiency of satellite brightness mitigation treatments (e.g., Tregloan-Reed et al. 2020, 2021, for Starlink’s Darksat), to space situational awareness, including the detection of unknown space debris.

We take advantage of the extended database of satellite observations conducted by the community of Unistellar telescope users through dedicated citizen science campaigns (e.g., Marchis et al. 2020). Unistellar’s network comprises a total of about 15 000 robotically-controllable telescopes of 11.4 cm aperture diameter, which are distributed worldwide and connected to ensure efficient data distribution. Thanks to a partnership with the SETI institute, Unistellar’s telescope network offers a unique chance to monitor the sky, including LEO satellites, from a wide range of Earth locations.

Our paper is structured as follows. In Sect. 2 we introduce the simulation procedure to construct the ground-truth data sets, and in Sect. 3 we describe the training of our two supervised neural networks. The results on the streak characterization performance are presented in Sect. 4. Finally, in Sect. 5, we discuss the ability of our pipeline to generalize to streaks detected with other types of telescopes, including deep images from 8-m class telescopes.

2. Data generation

In this section, we describe the procedure we followed to create the training datasets. **Given the moderate number and low diversity of real streaks detected with Unistellar telescopes, and given our objective of optimizing streak detection and segmentation as a function of (S/N), we favor training the neural networks on simulations with robust ground-truth streak properties.** We build a versatile simulation software to mock up realistic images with satellite streaks, for broad ranges of telescope types including Unistellar eVscope 1, eVscope 2, and Odyssey. We first present the overall simulation procedure in Sect. 2.1, followed by a description of the general properties of images from Unistellar telescopes in Sect. 2.2. We then summarize in Sect. 2.3 the properties of the simulated satellite streaks that will be used to train the neural networks.

2.1. Simulation procedure

The data generation process, described in detail in Janin-Potiron et al. (2024), is based on the *orbitAI* Python package. In this section, we introduce the three primary classes within this package: Telescope, Scene and Detector.

The Telescope unit is mainly responsible for image formation, with light propagation simulated using Fourier optics. The main controllable variables include the telescope’s diameter, focal length, observing band and pointing. Telescope’s pointing

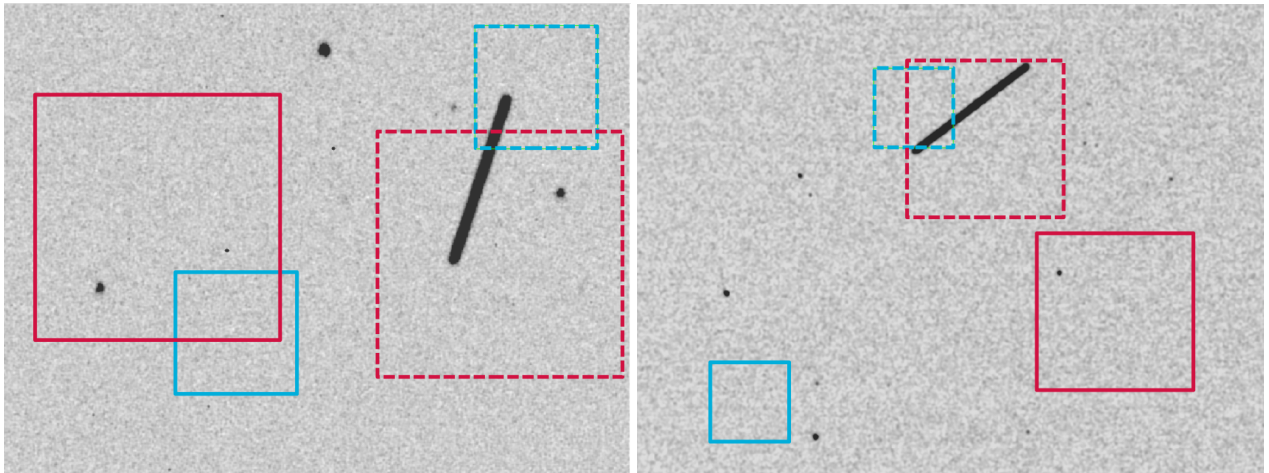


Fig. 1. Composite simulations of bright streaks created with the `orbitAI` package, for eVscope 1 (left) and eVscope 2 (right) image dimensions. The squares illustrate the random extraction of positive (dashed lines) and negative (solid lines) patches for training and validating the neural networks on small crops with and without satellite streak pixels. Patch sizes for the CNN and UNet are shown in blue and red, respectively. **With a $(S/N)_{\text{peak}} \approx 30$, these example streaks are on the high- (S/N) end of the data set distributions shown in Fig. 2. Streaks representative of the training sets are shown in Fig. 3.**

is managed using the `astropy WCS` (World Coordinate System) class, which facilitates the conversion between pixel coordinates and Alt/Az or RA/Dec coordinates. The effect of atmospheric turbulence on image formation is modeled by defining the telescope’s point-spread function (PSF) as a two-dimensional Gaussian, with a full width at half maximum (FWHM) corresponding to the Fried parameter r_0 . We account for streak intensity fluctuations (mainly due to satellite rotation and atmospheric scintillation) by generating random variations from a power spectral density derived from real on-sky images of bright satellites.

The `Scene` class is responsible for generating the observed scene, i.e. a star field with a satellite streak, as viewed through the telescope. There are three available subclasses for creating a `Scene`, each differing in how they generate the background image representing the star field behind the satellite streak: (1) `SimulatedScene`, where the positions and magnitudes of stars are randomly generated, (2) `CatalogScene`, where the positions and magnitudes of the stars are selected from catalogs (Hipparcos or Gaia at the moment) and (3) `CompositeScene`, where the simulated satellite streak is superimposed onto a real star field image captured by a Unistellar telescope. In this paper, we use only the `CompositeScene` mode, **which produces the most realistic simulations by retaining the distribution and diversity of astronomical sources, as well as the background properties and artefacts from real telescope images.**

The `Detector` unit is responsible for adding noise to the simulated satellite streak added to the background image. Since we exclusively use `CompositeScene`, the only noise that needs to be considered is photon noise, as readout and thermal noise are already present in the background image.

With a standard modern desktop computer, `orbitAI`, can generate 100 000 realistic simulations in under 5 hours while also keeping full control on the distributions of satellite altitude, orientation, and brightness in order to make the streaks representative of the target data set. By allowing efficient injection of synthetic streaks with realistic intensity fluctuations, and with accurate ground-truth mask bound-

Table 1. Parameter comparison for simulations in the high and low satellite flux regimes. $\mathcal{U}(a, b)$ stands for the uniform distribution between a and b .

	High flux regime	Low flux regime
Sat. Altitude (in km)	$\mathcal{U}(600, 900)$	$\mathcal{U}(600, 900)$
Streak mag.	$\mathcal{U}(5, 10)$	$\mathcal{U}(8, 11)$
Streak flux (in $\text{ph/m}^2/\text{s}$)	$[3.5 \times 10^5, 9 \times 10^7]$	$[3.5 \times 10^5, 5 \times 10^6]$
Streak length (in px)	$[150, 600]$	$[150, 600]$
r_0 (in cm)	$\mathcal{U}(2, 4.2)$	$\mathcal{U}(2, 4.2)$

aries, the software is well-suited to generate training sets for a broad range of astronomical surveys⁵.

2.2. Image properties

In this study, we mainly focus on images captured with Unistellar telescopes eVscope 1, eVscope 2, and Odyssey. The Unistellar satellite observation mode comprises reference frames with $t_{\text{exp}} = 4$ sec aimed at calibrating astrometry and photometry with reference stars, and a 1 min long sequence of short-exposure satellite frames with $t_{\text{exp}} = 150$ ms to record the motion of satellites over multiple exposures. This results in image depths (detection threshold) between 15 and 16 mag for stars in the $t_{\text{exp}} = 4$ sec frames, and around 10 mag for moving satellites in the $t_{\text{exp}} = 150$ ms frames. In this paper, we exclusively train and apply the neural networks on the short-exposure images, which have dimensions of 664×496 px, 1032×768 px, and 1464×1112 px for eVscope 1, eVscope 2, and Odyssey tele-

⁵ In contrast, when using real streaks instead of simulations, training sets are necessarily sparsely sampling the parameter space, and the definition of ground-truth masks at the pixel-level becomes much more challenging and time-consuming.

scopes, respectively. This corresponds to fields-of-view (FOV) of $\approx 19' \times 14'$ for eVscope 1, and $\approx 23' \times 17'$ for eVscope 2 and Odyssey, ensuring that streaks from LEO satellites are fully captured within the detector's field of view. Neural networks are trained on simulations from each of the three telescope versions in order to cope with their different pixel scales.

2.3. Simulated streaks properties

The simulated data sets were obtained from a total of 65 235 short-exposure images registered by amateur astronomers between January 2024 and February 2025, as part of 115 sequences of satellite observations with Unistellar telescopes. After flagging and excluding images with real streaks using SExtractor (Bertin & Arnouts 1996), we injected synthetic satellite streaks at random positions and orientation in the remaining images distributed over the 115 different fields-of-view. We produced two simulations per background image, to reach sufficient number of simulated streaks per telescope version. Together with the diverse satellite fluxes and streaks lengths, this ensure that individual simulations sample unique combinations of streak and background images properties. This strategy also helps build training sets with observing conditions and image quality representative of the target data set.

The signal-to-noise ratios (S/N) of injected satellite streaks are directly measured on simulations. The background noise levels σ_{sky} are measured over the entire fields-of-view of input telescope images (as opposed to patches, see Fig. 1), before streak injection. For each image, a Gaussian function is fitted to the pixel value distribution and we take σ_{sky} as the Gaussian standard deviation. We then use the simulated ground-truth satellite masks to identify pixels within the streaks, and to compute the S/N in three different ways. We use the maximum signal over pixels enclosed within the masks $(\text{S/N})_{\text{peak}}$, the average signal per pixel within the masks $(\text{S/N})_{\text{avg}}$, and finally the total signal within the satellite masks $(\text{S/N})_{\text{tot}}$. In this paper, performance are exclusively computed as a function of $(\text{S/N})_{\text{peak}}$.

We produced two different sets of simulations focusing either on the high or low-flux regime, to help stabilize the neural network performance over broader ranges in streak S/N. The properties of simulations forming these two data sets are summarized in Table 1 and shown in Fig. 2. **Representative examples are shown in Fig. 3.** In short, satellite altitudes and streak magnitudes are drawn from uniform distributions, resulting in different regimes of $(\text{S/N})_{\text{peak}}$ values. Given that significant numbers of streaks in the low-flux regime are below the background noise level, we further restrict this data set to faint, but detected streaks with $(\text{S/N})_{\text{avg}} > 0.2$ and $(\text{S/N})_{\text{peak}} > 2$. This results in a drop of the number of streaks with $\text{mag} > 9$ in the bottom-right panel of Fig. 2.

3. Deep learning pipeline

In this section, we describe the development and training of the two-step deep learning pipeline aimed at analyzing astronomical images affected by satellite streaks. After introducing the data preprocessing in Sect. 3.1, we introduce in Sect. 3.2 the convolutional neural networks to identify individual images containing streak candidates among large, **unbalanced data sets with high fractions of non-contaminated images**. In Sect. 3.3, we then present neural networks dedicated to the pixel-level reconstruction of streaks (UNet) under various conditions and S/N. **In Sect. 3.4, we then present the total inference times for realistic**

data sets, in order to illustrate the gain obtained with our two-step approach.

3.1. Image preprocessing

To enable the flexible classification and segmentation of large telescope images of arbitrary dimensions, we trained the neural networks on square patches extracted from the simulated images. For each simulation, we randomly extracted two patches; one "positive" patch covering the inject satellite streak, and one "negative" patch without any pixel affected by the streak. Positive patches were required to include a minimum number of pixels within the satellite mask, min_{pp} . After extracting various sets of patches with min_{pp} ranging between 10 and 500 pixels, we obtained slightly better performance for CNNs and UNets trained with $\text{min}_{\text{pp}} = 100$ and kept this value as baseline. We also compared CNNs and UNets trained on patch sizes ranging from 32×32 to 256×256 pixels following the procedure described in Sect. 4. Default patch sizes of 128×128 and 256×256 pixels were adopted for training the CNNs and UNets, respectively, and their dimensions relative to the overall images are illustrated in Fig. 1. The models were then applied to entire images using sliding windows, as further described in Sect. 4.

Neural networks are not only training more efficiently on data sets with homogeneous range of values and contrasts, but their application also requires similar dynamic ranges for values in the target data sets. Using a local normalization of pixel values in patches from the training, validation, and test sets allows to circumvent these issues while retaining the relevant morphological information. As baseline, we followed Paillassa et al. (2020) and applied an arcsinh transformation to each patch D , in order to reduce the contrast between small and large pixel values:

$$\tilde{D} = \text{arcsinh} \left(\frac{D - \mu_{\text{patch}}}{\sigma_{\text{patch}}} \right), \quad (1)$$

where μ_{patch} and σ_{patch} are the local average and standard deviation of pixel values within the entire patch. Other local methods have been tested such as a min-max normalization, scaling to the $[-1;1]$ range, normalizing to zero mean and unit variance, or applying a square-root stretch instead of the arcsinh. We also tried using μ_{img} and σ_{img} measured over the larger telescope images, and using $\mu_{\text{patch, corners}}$ and $\sigma_{\text{patch, corners}}$ computed over 10×10 pixel corners without contamination from the satellite streaks. None of these approaches improved the overall network performance compared to the baseline.

3.2. Image classification

3.2.1. Training the CNN

The CNNs for binary classification of patches with or without streaks are inspired from the AlexNet architecture (Krizhevsky et al. 2012). After the input $128 \times 128 \times 1$ layer, the baseline model includes four convolutional layers with kernel sizes of 9×9 , 5×5 , 5×5 , 3×3 , and with 16, 32, 64, 128 filters, respectively, and three fully-connected (FC) layers with 2048, 256, and 16 neurons, respectively (see Appendix A). Rectified Linear Unit (ReLU) activation functions are used between each of these layers. Max-pooling layers with 2×2 kernels and stride = 2 are inserted between the convolutional layers to reduce the number of network parameters, and help learn small translation invariance. The output layer is a single-neuron layer with sigmoid activation, and results in output scores $p_{\text{CNN},l}$ scores per patch l ranging between 0 (no streak) and 1 (streak candidate).

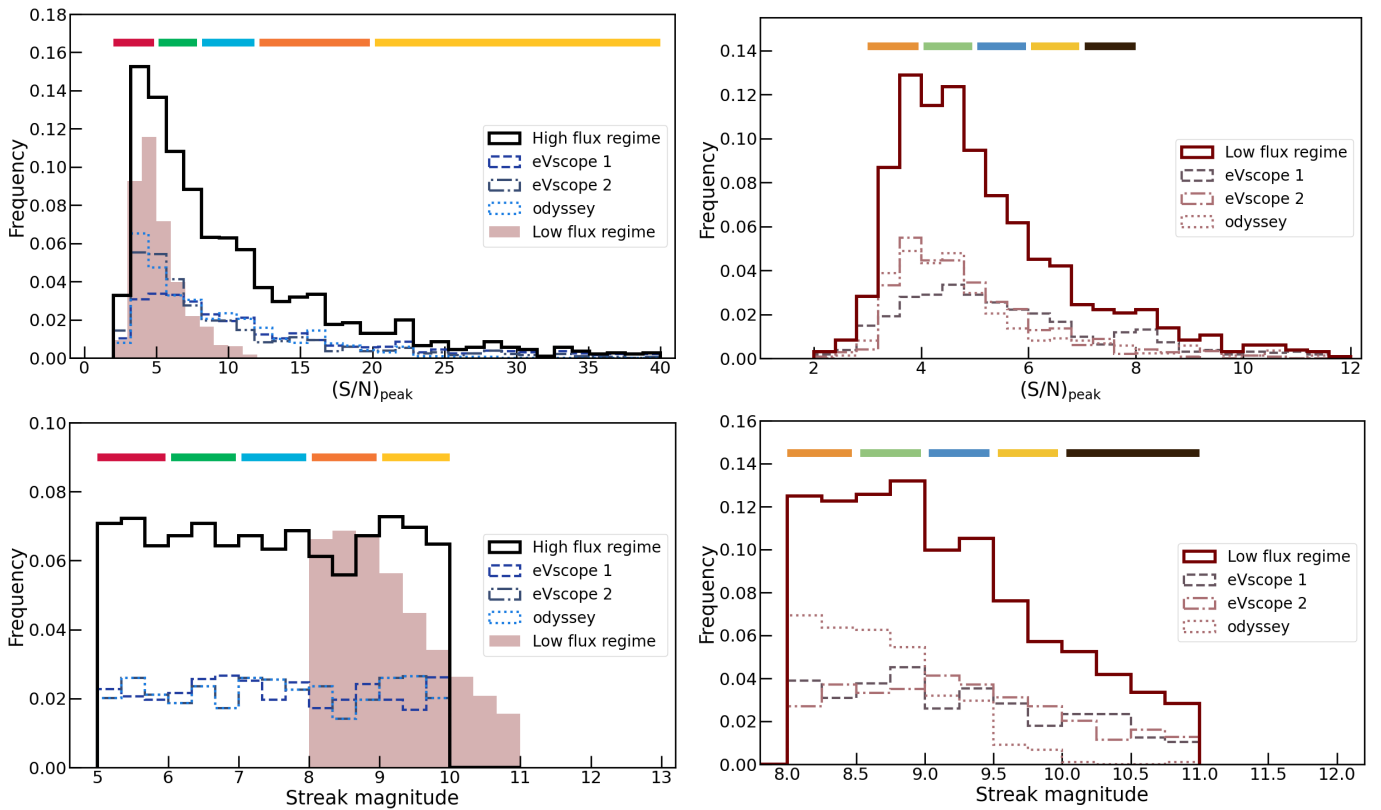


Fig. 2. Distributions of $(S/N)_{\text{peak}}$ (top panels) and magnitudes (bottom panels) for simulated satellite streaks belonging to the high-flux (left) and low-flux (right) regime data sets. Horizontal bars mark the different bins to measure the evolution of network performance in Sect. 4.

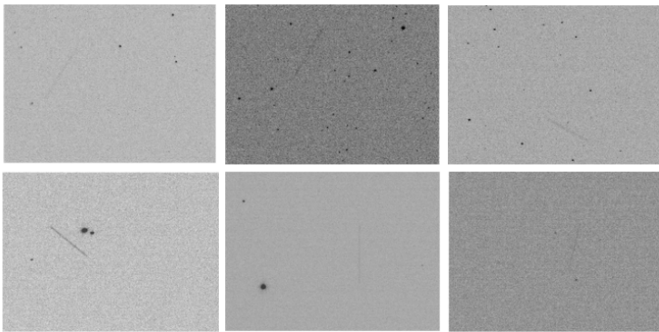


Fig. 3. Examples of streaks from the high-flux dataset injected with orbitAI. Most of these examples have a $(S/N)_{\text{peak}} \lesssim 7$ and $(S/N)_{\text{avg}} \lesssim 1$. A majority of pixels within the ground-truth masks therefore have a streak signal below the image background noise level, which complicates the mask reconstruction at the pixel-level.

We tested different architectures adapted from this baseline by varying the number of hidden layers. For each model, we optimized the kernel sizes and number of filters per convolutional layer, and the number of neurons per FC layer using a grid search. We also explored the influence of the kernel sizes in the max-pooling layers, and the use of dropout regularization (Srivastava et al. 2014) with a dropout rate of 0.5 between the FC layers. We obtained classification performance comparable to the baseline CNN for a number of convolutional layers ranging between two and four, showing that simple CNNs are sufficient to learn the simple signatures from satellite streaks. For

this reason, deeper models with residual connections (ResNet, He et al. 2015) were not explored.

For training and validating the CNN classifier, we used a subset of 25 000 simulations in the high flux regime, with uniform streak magnitude distribution between 5 and 10 mag, and equal number of images from eVscope 1, eVscope 2, and Odyssey telescopes. We extracted a balanced data set with 50% positive patches and 50% negative patches, and kept 36 000 and 9 000 patches for training and validation, respectively. Optimization of the network parameters was achieved by minimizing the binary cross-entropy loss expressed as:

$$\text{BCE}(y, p_{\text{CNN},i}) = -\frac{1}{N} \sum_{k=1}^N y_k \log(p_k) + (1 - y_k) \log(1 - p_k), \quad (2)$$

where y_k and p_k are the values of the ground-truth and predicted labels respectively. Values of p_k result from the sigmoid activation and range from 0 to 1. We trained the networks with the Adam optimizer (Kingma & Ba 2017), using a number of images per batch N of 32 and a learning rate fixed to 0.0001, after optimizing these hyperparameters with a grid search. The CNNs were trained over a total of 100 epochs and the final models were saved at the epoch showing the lowest binary cross-entropy loss in the validation set.

In addition, we trained a three-class variant of the CNN classifier in order to separate high- and low- S/N streak candidates and conduct the pixel-level reconstruction of satellite masks with the adequate network. This variant shares the same architecture as the binary classifier, with three neurons in the output layer instead of one. This network was trained with the same set of 25 000 simulations, using patches with $(S/N)_{\text{peak}} > 8$ as examples of bright streaks (class 1), patches with $(S/N)_{\text{peak}} < 8$ as

faint streaks (class 2), and the remaining patches without injected streak as negatives (class 3).

3.2.2. Application to larger images

To classify the larger telescope images of variable dimensions, we applied the CNNs to sliding windows of 128×128 pixels, and combined the resulting $p_{\text{CNN},l}$ scores per sliding window l into a prediction per image p_{img} . We excluded a five-pixel wide contour to avoid introducing classification biases due to the artefacts around image borders. We used a 50-pixel step between sliding windows to keep a significant spatial overlap and mitigate the impact on streaks located near window edges. Smaller steps did not show a strong improvement and increased the inference time. For a given image, the CNN scores were computed independently for each sliding window, and we computed the minimum, maximum, average, and percentiles of the overall $p_{\text{CNN},l}$ score distribution. Better performance were obtained by assigning the 99.9% percentile of the distribution to the score of the entire image p_{img} .

Various processing methods were implemented, such as cleaning hot pixels with median filtering, and applying Gaussian smoothing or spatial binning of the entire images. We also tried to group the scores from neighboring windows, to account for the longer extension of simulated streaks compared to the window size. We attempted 2×2 , 3×3 , and 4×4 binning of the two-dimensional $p_{\text{CNN},l}$ score array, by taking either the mean or the median of scores per bin. None of these approaches improved the classification performance with respect to the baseline.

3.3. Segmentation of satellite streaks

To identify the pixels affected by satellite streaks, we trained a fully-convolutional neural network (UNet) to predict segmentation maps ("satellite masks" hereafter) with pixels belonging to satellite streaks labeled as 1, and other pixels distributed over the sky background, stars, or other sources labeled as 0.

3.3.1. Training the neural networks

Our deep-learning framework is based on a U-shaped neural network adapted from the original UNet from Ronneberger et al. (2015). Such U-shaped architectures are divided into encoder and decoder components with skip connections inbetween. The encoder learns the morphological features in the input images at various scales, using a sequence of convolutional layers with downsampling. Based on the encoded features, the decoder combines upsampling and concatenation to restore the information extracted at different scales, and predict an output segmentation mask with the same resolution as the input image. UNet have become state-of-the-art for astronomical image segmentation tasks, and they cover broad ranges of applications from the segmentation and deblending of galaxies (e.g., Boucaud et al. 2020), to the identification of filamentary structures over the Galactic plane (Zavagno et al. 2023; Berthelot et al. 2024), or to searches of strong gravitational lenses (e.g., Zhong et al. 2024). Since the original UNet is optimized for analyses of biomedical images with complex morphological structures, we used shallower network architectures inspired from the cosmic ray segmentation pipelines developed in Zhang & Bloom (2020) and Xu et al. (2023) for space- and ground-based astronomical images.

Our baseline UNet starts from an input layer of 256×256 pixels, followed by an encoder with two blocks comprising two

convolutional layers with 3×3 kernels and ReLU activation per block. Between these block, max-pooling layers with 2×2 kernels allows to decrease the dimension of features maps from 256×256 in the first block to 128×128 in the second block, and 64×64 in the network bottleneck. Starting from the coefficients of the 128 filters in the bottleneck, the decoder performs transposed convolutions with stride of 2, and concatenates the resulting filters with the skip connections from the corresponding block of the encoder, in order to propagate relevant information at all scales. The upsampling blocks are followed by two convolutional layers with 3×3 kernels and ReLU activation. The output layer is a 1×1 convolution layer with sigmoid activation and results in the output satellite mask with scores per pixels ranging from 0 (pixel outside streak) to 1 (pixel within streak). This baseline architecture is illustrated in Appendix A.

As for the CNN, we tested several variations around this baseline UNet. We successively added and removed one block in the encoder and in the decoder, in order to test deeper and shallower network architectures. Using a single downsampling and upsampling stage prevented the UNet to simultaneously learn the low- and high-level features, resulting in poor streak reconstruction. In contrast, increasing network depth to three downsampling and three upsampling stages did not significantly improve the output segmentation compared to the baseline. After exploring the influence of the number of filters per convolutional layer, we used a fixed number of 32, 64, and 128 filters in the first block, second block, and bottleneck, respectively. In addition, we explored using average pooling instead of max-pooling layers in the encoder, using upscaling layers in the decoder instead of transposed convolutions, and discarding the skip connections. Discarding the skip connections proved helpful in some cases to put more focus on the large-scale features during training, and predict smoother satellite masks. Such UNet are referred to as UNet_{noskip} in Sect. 4.

Other alternative architectures were implemented, including the original UNet from Ronneberger et al. (2015), the UNet from Zhang & Bloom (2020), and the UNet++ from Zhou et al. (2018). The UNet++ replaces the skip connections from the UNet with skip pathways combining convolutional layers and dense skip connections, in order to better capture multi-scale information and improve the overall segmentation accuracy (Zhou et al. 2018). Among these three tests, only adapting the UNet++ to the same depth as our baseline UNet improved the performance in some regimes. This network is discussed in Sect. 4 and referred to as UNet_{++mod}. Lastly, we tested replacing the local normalisation of patches described in Sect. 3.1 with a preprocessing layer implemented directly within the UNet between the input layer and the first convolutional layer, as previously done by Richards et al. (2024) for their segmentation of low-surface brightness Galactic structures. This layer automatically scales the intensity of image X as $X_{\text{scaled}} = \text{arcsinh}(aX + b)$, where coefficient a and b are optimized during network training. This strategy provided similar satellite mask predictions as the baseline UNet trained on normalized patches.

To explore the streak reconstruction performance over broad ranges of (S/N), we trained the baseline UNet and each alternative architecture using the simulated data sets in the high-flux (UNet_{high}) and low-flux (UNet_{low}) regimes. In both cases, we used balanced data sets with 50% positive patches and 50% negative patches, and equal number of patches extracted from eVscope 1, eVscope 2, and Odyssey images. For UNet_{high} (UNet_{low}), this corresponds to 27 072 (36 000) and 6768 (9000) examples for training and validation, respectively. We trained the networks up to 100 epochs with the Adam optimizer, a batch size of 32,

and using the binary cross-entropy loss per pixel to penalize wrong predictions:

$$\text{BCE} = -\frac{1}{N} \sum_{k=1}^N \sum_{j=1}^Y \sum_{i=1}^X y_{i,j,k} \log(p_{i,j,k}) + (1 - y_{i,j,k}) \log(1 - p_{i,j,k}), \quad (3)$$

where $y_{i,j,k}$ and $p_{i,j,k}$ are the values of the ground-truth and predicted satellite masks for pixel (i,j) from image k , respectively, and X , Y , and N , are the image dimensions in pixel units and the number of images per batch, respectively. In contrast to the CNN, we implemented a learning rate scheduler rather than using a fixed learning rate. We decreased the learning rate by a factor 10 every five epochs. The initial learning rate at the first epoch was optimized with a grid search and fixed to a value of 0.01. The scheduler enable a quicker convergence of the training and validation loss curves compared to a constant learning rate, and resulted in moderate improvement of the final performance.

3.3.2. Prediction of satellite masks

The UNet trained on 256×256 pixel patches were applied to the entire images using overlapping sliding windows. We extracted a given patch from the telescope image, applied a local normalization of pixel values as in Sect. 3.1, and inferred the neural network predictions before moving on to the next patch. Nearby, overlapping patches may have different segmentation due to the local normalization. We combined the results by taking the average UNet output score per pixel, in order to obtain the final satellite masks over the entire images. Averaging the score per pixel obtained from multiple overlapping sliding windows allowed us to minimize the artefacts resulting from variations in the UNet predictions per patch, and to obtain smooth satellite masks.

While such UNet can be applied to arbitrary window sizes, optimal performance was obtained for sliding windows of dimensions identical to the training data (256×256 pixels). The accuracy of predicted masks smoothly increases when decreasing the step size between windows, at the expense of higher computational costs. As tradeoff, we used steps of 10 pixels between each sliding window. The ten-pixel wide band with a single UNet prediction per pixel was discarded around image borders.

3.4. Inference time

Image processing times were evaluated on a single CPU, as we intend to simplify the neural network implementation into existing data processing pipelines. As an example, we took a typical Unistellar data set with 380 short-exposure images of 1032×768 px, including 5 images with a LEO streak and 375 non-contaminated images. The CNN classifier requires a total CPU time of 0.71 sec to process each image, and the UNet requires 65 sec per image for 10-pixel steps between sliding windows. These estimates include image loading, patch preparation and pre-processing, model evaluation and the final conversion between score per patch to score per image (or per pixel for the UNet). With our two-step approach, we therefore need a total of 10 min to process an entire Unistellar sequence.

Conducting both the streak detection and segmentation with a single UNet would require between 5 and 7 hours for this test sequence, depending on the number of patches grouped into the same batch during network evaluation. One way to speed up the total UNet inference time is to use larger

Table 2. Content of the various test sets (see also Sect. 4.1).

Test set	TS _{obs}	TS _{hybrid}	TS _{sim,high}	TS _{sim,low}
for CNN	✓	✓		
for UNet			✓	✓
Set	426 pos.	2008 pos.	2008 sims	1312 sims
Content	25967 neg.	25967 neg.	with streak	with streak
Balanced tel. version		✓	✓	✓
Streak brightness	(S/N) _{tot} ∈ [200; 5e4]	as Fig. 2 (left)	as Fig. 2 (left)	as Fig. 2 (right)

pixel steps between patches, which reduces the number of network evaluations per pixel. For the Unistellar image quality, this strategy is however infeasible as it introduces several artefacts near the patch and image boundaries, which are then identified as false-positive streak detections.

This shows the benefit of using a two-step approach, with a shallow CNN conducting rapid image classification, and excluding the larger fraction of non-contaminated astronomical images before attempting a mask reconstruction. Based on Unistellar campaigns targetting random sky positions, we estimate that about 3% of 60 sec exposures from imaging surveys contain a LEO streak in 2025 (for random observing dates and fields-of-view of $23' \times 17'$). While this ratio is expected to increase in the coming years, it remains at the percent level, which further motivates the implementation of a binary classifier (as also proposed by Liu et al. 2025).

4. Evaluation of the network performance

4.1. Construction of the test sets

The performance of the CNN classifiers were evaluated using multiple test sets including either only real observations, or a mix of real and simulated images. Firstly, we assembled a test set with 75 sequences of satellite observations collected between January and September 2024 (TS_{obs}), distinct from the 115 sequences used for training and validation. These observations were inspected, both visually and with SExtractor, to label the exposures with and without a satellite streak. We identified a total of 426 positive examples with a detected streak, and 25 967 negative examples over these 75 sequences. About 3 to 10 exposures per sequence of 400 images contain a streak, either from the primary target, or from a secondary satellite crossing the field-of-view. About 70%, 20%, and 10% of images in this test set were taken with eVscope 1, eVscope 2, and Odyssey telescopes, respectively. These fields cover a wide range of stellar densities and the 75 sequences have variable imaging quality, including field-of-view drifts and other types of artefacts that are representative of the target data set. The (S/N)_{tot} distribution of real streaks in set TS_{obs} range from 200 to 50 000, with a peak at (S/N)_{tot} of 9000. The faintest detections were obtained during the Advanced Composite Solar Sail System (ACS3) observing campaign of August 2024. Secondly, a separate set (TS_{hybrid}) was created to further test on fainter satellites with lower S/N, and to probe the evolution of performance as a function of the ground-truth streak properties. This set combines the 25 967 real images without streak as negative examples, with 2008 simulations randomly selected from the 75 fields-of-view as positive examples. TS_{hybrid} includes a balanced fraction of eVscope 1, eVscope 2, and Odyssey images, in order to compare performance as a function of the detector version.

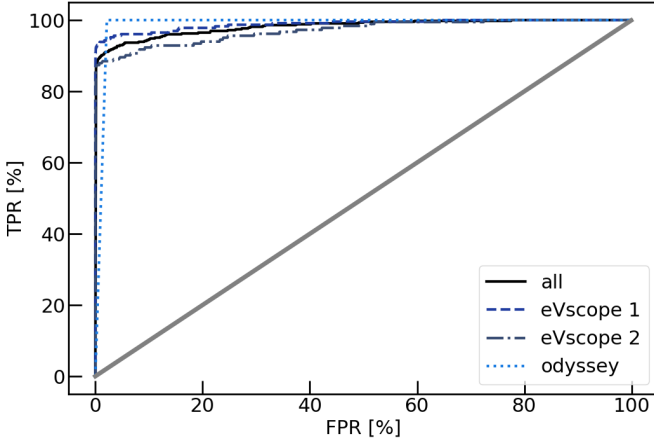


Fig. 4. Receiver Operating Characteristic curves (ROC) showing the performance of the CNN classifier on test set TS_{obs} with 426 real streaks as positive examples and 25 967 negative examples (black solid curve). The blue dashed, dash-dotted, and dotted lines distinguish the ROC curves from eVscope 1, eVscope 2, and Odyssey observing sequences part of TS_{obs} , respectively. The scores per image p_{img} are obtained from the 99.9% percentile of the distribution of scores per sliding window $p_{CNN,1}$. For reference, the grey line indicates a random classifier.

The performance of the UNet were exclusively evaluated on simulated images, due to the difficult extraction of robust ground-truth satellite masks from real observations. We used simulations randomly selected from the same parent data set as the training and validation sets, taking 10% of the simulations produced per observing sequence to preserve a large variety of fields-of-view. We also required an equal number of eVscope 1, eVscope 2, and Odyssey images. As a result, UNet_{high} and UNet_{low} were tested on set $TS_{sim,high}$ and $TS_{sim,low}$ with 2008 and 1312 images, respectively. The properties of streaks in these test sets follow the distributions shown in Fig. 2.

4.2. Performance metrics

The performance of neural networks were compared using the classical metrics for binary classification problems. For the CNN classifiers, we plotted the receiver operating characteristic (ROC) curves between the true positive rate (TPR or recall) and the false positive rate (FPR), which are defined as:

$$TPR = \frac{TP}{TP + FN}, \quad FPR = \frac{FP}{FP + TN} \quad (4)$$

where TP, FP, TN, and FN refer to the number of true positives (images with streak correctly identified), false positives (images with streak candidate wrongly identified by the CNN), true negatives, and false negatives, respectively. We computed the area under ROC curve (AUC) to compare performance and select networks having the highest TPR at fixed FPR, corresponding to AUC approaching the ideal value of 1. Moreover, we optimized the recall at low contamination using the TPR at a FPR of 1% (TPR_1 hereafter) to ensure that the classifier can extract the rare candidates streaks from large, unbalanced data sets. For the three-class CNN, we used the same metrics as for the two-class scenario, and compared a given class with the other two (one-vs-rest strategy, e.g., Bishop 2006).

For the segmentation stage, we also relied on the ROC curves to compare the satellite mask predictions from various UNet. We

quantified the TPR (or recall) and the FPR at the pixel level, by summing over pixels and over images in the test set, as follows:

$$TPR = \frac{\text{number of streak pixels correctly classified}}{\text{total number of ground truth streak pixels}}, \quad (5)$$

$$FPR = \frac{\text{number of nonstreak pixels incorrectly classified}}{\text{total number of ground truth nonstreak pixels}}. \quad (6)$$

We optimized the recall at a very low FPR of 0.001% ($TPR_{0.001}$ hereafter) in order to achieve the reconstruction of accurate satellite masks with a contamination level of maximum 10 false positive pixels per image. The AUC and $TPR_{0.001}$ values were combined with the precision defined as:

$$\text{precision} = \frac{\text{number of streak pixels correctly classified}}{\text{number of predicted streak pixels}}. \quad (7)$$

The precision metric is only sensitive to positive (streak) pixels, and is not affected by the imbalanced and variable ratio of streak over non-streak pixels in real images. Using the precision-recall curves, we quantified the precision of the satellite masks predicted by the UNet, for a fixed threshold in TPR of 90% (P_{90} hereafter).

To assess the ability of the UNet to predict accurate elongations of satellite streaks, we complemented these metrics with a measurement of the offset between the endpoint coordinates from the predicted and ground-truth satellite masks. Characterizing streak lengths is key to determine the ability of the pipeline to constrain the movement of streaks across the sequences of short exposures, as required for potential space surveillance applications. For this purpose, the satellite masks predicted by the UNet (see Sect. 3.3.2) were binarized and the handful of compact, spurious false-positive detections over the frames were masked out. We then summed up the offset between predicted and ground-truth endpoint coordinates at both streak ends, ignoring the possible discontinuities along the binarized masks, and we normalized this offset by the total streak lengths in pixel units. Hyperparameters involved in the creation of the final masks were optimized, and we adopted the value of $p_{UNet} = 0.5$ and a minimum number of 64 contiguous pixels per source in the satellite masks. Masks with predicted endpoint coordinates offset by $> 10\%$ of the total streak length were referred to as “catastrophic failures”.

4.3. Performance of the CNN

The performance of the baseline CNN classifier on the test set of real satellite observations TS_{obs} is shown in Fig. 4. We obtain an AUC of 0.9817 and a TPR at 1% FPR of 89.9%, showing that the network can efficiently identify the rare satellite streaks from a large number of images. A recall of nearly 100% is reached at the expense of higher contamination rates of $\approx 20\%$. This illustrates the good generalization capabilities of the CNN to diverse observing conditions and imaging quality. Moreover, the high recall allows to efficiently flag images with candidate streaks for subsequent analysis with the segmentation networks. Fig. 4 also show that ROC curves remain stable for sequences taken with different telescope types, with AUC of 0.9885, 0.9705, 0.9893 for eVscope 1, eVscope 2, and Odyssey, respectively.

The predicted image classifications were visual inspected for a fiducial score threshold corresponding to 1% FPR. We noticed that images with bright satellite streaks, and pointing instabilities causing non-linear streak signatures are all classified correctly as true positives. Images with extended stellar light profiles similar

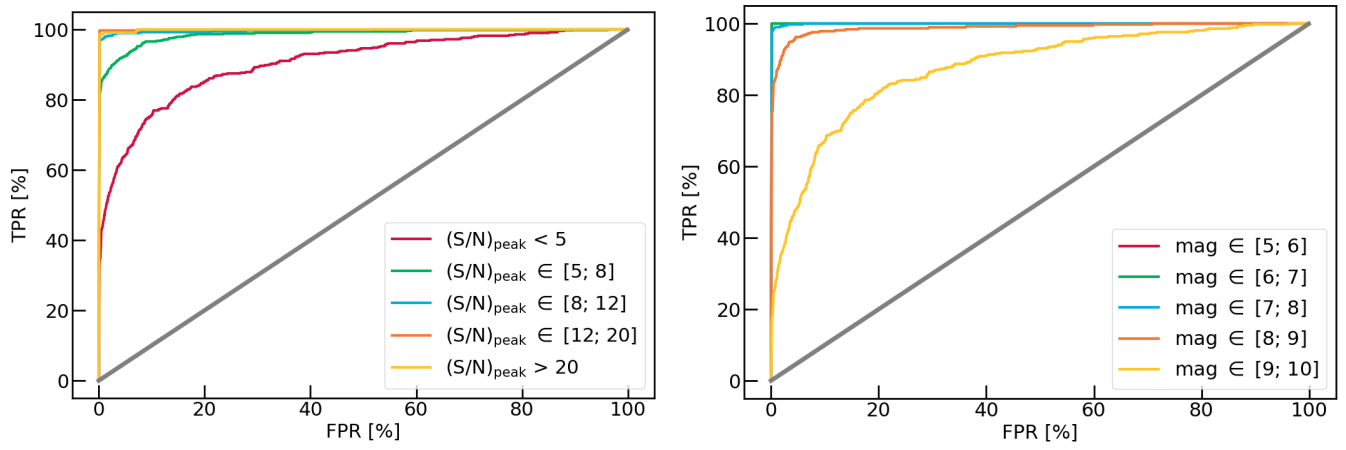


Fig. 5. *Left:* ROC curves of the CNN classifier for different bins of streak $(S/N)_{\text{peak}}$, using $\text{TS}_{\text{hybrid}}$ with 2008 simulations as positive examples, and 25 967 real images without streak as negative examples. *Right:* Evolution of the performance as a function of the streak magnitude. The scores per image p_{img} are obtained from the 99.9% percentile of the distribution of scores per sliding window $p_{\text{CNN},1}$. For reference, the grey line indicates a random classifier.

Table 3. Evaluation of the performance of $\text{UNet}_{\text{high}}$ for segmentation of satellite streaks in set $\text{TS}_{\text{sim,high}}$.

Test set	area under ROC	TPR at 0.001% FPR	precision at 90% recall
all	0.9889	84.9%	97.8%
eVscope 1	0.9986	89.8%	99.9%
eVscope 2	0.9952	84.6%	98.5%
Odyssey	0.9833	84.9%	96.5%
$(S/N)_{\text{peak}} < 5$	0.9344	35.2%	0.1%
$(S/N)_{\text{peak}} \in [5; 8]$	0.9896	80.7%	95.8%
$(S/N)_{\text{peak}} \in [8; 12]$	0.9978	92.6%	99.9%
$(S/N)_{\text{peak}} \in [12; 20]$	0.9995	96.9%	100.0%
$(S/N)_{\text{peak}} > 20$	1.0	97.7%	100%
$\text{mag} \in [5; 6]$	1.0	98.2%	100.0%
$\text{mag} \in [6; 7]$	1.0	95.0%	100.0%
$\text{mag} \in [7; 8]$	0.9999	87.0%	99.3%
$\text{mag} \in [8; 9]$	0.9970	52.2%	76.6%
$\text{mag} \in [9; 10]$	0.8313	9.7%	1.5%

The metrics evaluated on TS_{obs} and $\text{TS}_{\text{hybrid}}$ demonstrate that the CNN classifier can efficiently identify images with streak candidates among real observing sequences. With the optimal score threshold of $p_{\text{img}} = 0.98$ that maximizes TPR_1 over set TS_{obs} , one can select images to be further analysed with the UNet. With this score threshold, for each sequence in set TS_{obs} , at least one image with a detected streak is recovered. This suggests that the false-negative rate is low enough to ensure that the CNN correctly flags sequences with a LEO satellite, even though some individual images with faint, partial streaks might be discarded. In addition, with $p_{\text{img}} = 0.98$ corresponding to a FPR of 1%, for a typical sequence with 400 short-exposure images including 4 (396) positives (negatives), we expect 3–4 images without streak incorrectly classified as positives. These false positives can be easily excluded after UNet segmentation, as their predicted satellite masks typically have small numbers of positive pixels (see Sect. 4.4).

Finally, our three-class variant of the CNN classifier can be used to redirect the bright streak candidates to $\text{UNet}_{\text{high}}$, and the fainter streaks to UNet_{low} , in order to ensure optimal reconstruction of streaks in both S/N regimes.

4.4. Performance of the UNet

4.4.1. High flux regime

Training the various UNet architectures from Sect. 3.3.1 on high-flux simulations did not provide major differences. In consequence, performance are only evaluated for the baseline UNet, which we refer to as $\text{UNet}_{\text{high}}$, using set $\text{TS}_{\text{sim,high}}$. Results are summarized in Table 3 and Fig. 6, where we combined all simulations, separated the three telescope versions, and measured the evolution as a function of streak $(S/N)_{\text{peak}}$ and total magnitude.

The ROC and precision-recall curves in Fig. 6, and the AUC, precision and recall at low contamination listed in Table 3 demonstrate the optimal performance for the reconstruction of simulated streaks from $\text{TS}_{\text{sim,high}}$. We obtain a global AUC of 0.9889, $\text{TPR}_{0.001}$ of 84.9%, and P_{90} of 97.8%, showing the overall good fidelity of predicted masks, despite the inclusion of relatively faint streaks with $(S/N)_{\text{peak}} < 5$ in $\text{TS}_{\text{sim,high}}$ (see Fig. 2).

The evolution of metrics as a function of streak properties show that the best reconstructions of satellite masks are ob-

to satellite streaks are also correctly labeled as true negatives. On the other hand, most false-negatives correspond to exposures with satellites crossing the image borders, and leaving incomplete morphological signatures. Excluding these images from set TS_{obs} would improve the recall at 1% FPR to nearly 99%.

The performance of the CNN classifier were evaluated as a function of the ground-truth streak properties using set $\text{TS}_{\text{hybrid}}$, with 2008 simulations as positive examples, and 25 967 real images without streak as negative examples. The ROC curves for various streak magnitude bins are plotted in Fig. 5 and indicate nearly ideal performance up to $\text{mag} \approx 8$, and substantial degradation at $\text{mag} \geq 9$. TPR_1 decreases from 99–100% at $\text{mag} \approx 8$, to 86.8% for the $\text{mag} \in [8; 9]$ bin, and 28.6% for the $\text{mag} \in [9; 10]$ bin. TPR_1 is comparable for the three telescope versions at $\text{mag} \leq 8$, and slightly inferior for eVscope 2 at $\text{mag} \geq 9$. In terms of $(S/N)_{\text{peak}}$, the performance are optimal for $(S/N)_{\text{peak}} > 12$, with AUC of 0.999 and TPR_1 of 99–100%. For fainter streaks with $(S/N)_{\text{peak}} \in [8; 12]$, $\in [5; 8]$, and < 5 , the AUC is 0.9963, 0.9852, and 0.9050, and TPR_1 is 97.7%, 86.5%, and 47.1% respectively.

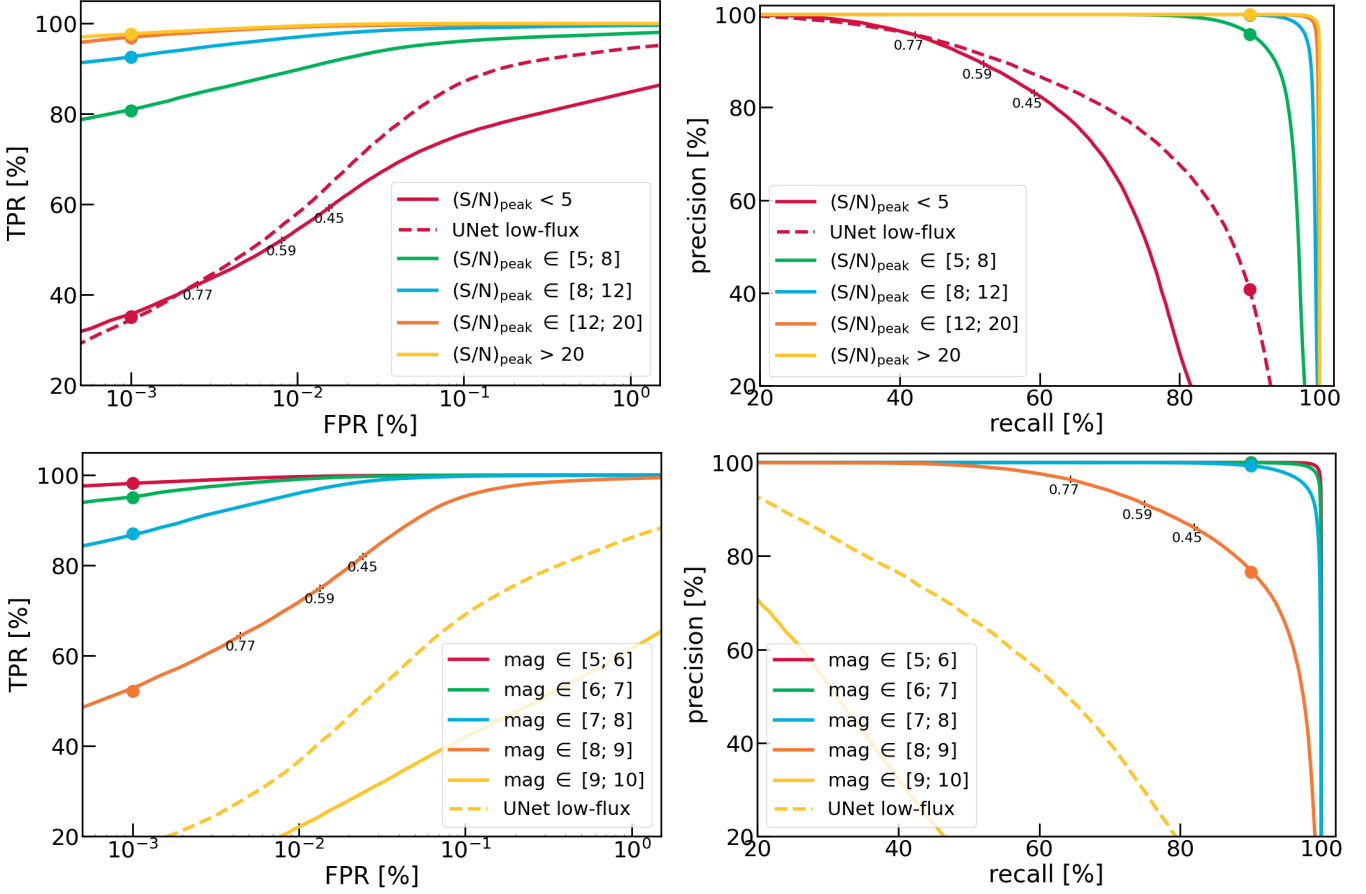


Fig. 6. Performance of UNet_{high} evaluated on set TS_{sim,high}, as a function of streak properties divided into $(S/N)_{\text{peak}}$ bins (*top*), and magnitude bins (*bottom*). *Left*: ROC curves with large dots marking the TPR levels at 0.001% FPR ($\text{TPR}_{0.001}$). *Right*: Precision-recall curves with dots showing the precision at 90% recall (P_{90}). In all panels, the performance of UNet_{high} on the faintest streaks with $(S/N)_{\text{peak}} < 5$ or $\text{mag} > 9$ (solid curves) is compared with UNet_{low} (dashed curves).

tained for $(S/N)_{\text{peak}} > 8$, with $\text{TPR}_{0.001} \geq 93\%$, and $P_{90} \approx 100\%$. The predictions gradually degrade for fainter streaks with $(S/N)_{\text{peak}} \in [5; 8]$, with values of 80.7% and 95.8% for $\text{TPR}_{0.001}$ and P_{90} , respectively. The last bin with $(S/N)_{\text{peak}} < 5$ shows the limitations of UNet_{high} for the segmentation of the faintest streaks partially below the background noise levels. In that bin, only 35.2% of pixels within the ground-truth satellite masks are correctly classified, for a FPR of 0.001%, and the precision only reaches values $> 80\%$ for a recall lower than 60% (top-right panel of Fig. 6). The evolution with streak magnitude suggests similar trends, with optimal segmentation at $\text{mag} \leq 8$, and a 23% drop in P_{90} for $\text{mag} \in [8; 9]$. The UNet produces unreliable masks for $\text{mag} \geq 9$, despite an overall AUC > 0.8 in that bin. **This very low- (S/N) regime produces typical failure cases, where the extremely low signals prevent UNet_{high} from reconstructing complete satellite masks. This regime is the focus of UNet_{low} in Sect. 4.4.2.**

Table 3 indicates small, but non-negligible differences in performance as a function of telescope type (see also Appendix B). The values of $\text{TPR}_{0.001}$ and P_{90} increase by 4.9% and 3.4%, respectively, between test Odyssey and eVscope 1 simulations. This effect results from differences in the S/N of eVscope 1, eVscope 2, and Odyssey simulations, which have brightnesses drawn from uniform magnitude distributions, but different pixel sizes. As seen in Fig. 2, fewer eVscope 1 images have $(S/N)_{\text{peak}} \approx 4-5$, and their overall $(S/N)_{\text{peak}}$ distribution is squeezed to higher

values. As a consequence, $\text{TPR}_{0.001}$ averaged over all eVscope 1 images in TS_{sim,high} increases. The impact of variable S/N distributions for eVscope 1, eVscope 2, and Odyssey is further assessed in Appendix B.

To quantify the impact of blending, we compared the performance of UNet_{high} between isolated streaks, and streaks overlapping other sources in the field. For each simulation in TS_{sim,high}, we used SExtractor to extract sources from the background Unistellar images (before streak injection), and we searched for spatial overlap between detected stars (at 2σ) and the injected streaks. We obtained 103 (1905) blended (nonblended) streaks in TS_{sim,high}, allowing us to compute the performance over both subset, for the five $(S/N)_{\text{peak}}$ bins. A comparison with the metrics listed in Table 3 shows that overlapping sources only produce minor degradations in performance, essentially at the pixel level. For instance at $(S/N)_{\text{peak}} > 20$, $\text{TPR}_{0.001}$ only decreases from 97.7% to 96.0% between the isolated and blended subsets. At $(S/N)_{\text{peak}} \in [5; 8]$, $\text{TPR}_{0.001}$ decreases from 81.1% to 71.9%. While these results are affected by low-number statistics in the blended subset, examples shown in Appendix D indicate that blending does not drastically affect the output mask shapes when the streak and nearby stars have comparable peak (S/N). However, when the stellar over streak luminosity ratio increases, UNet_{high} tends to either blend or mask out both components. This is a typical failure case.

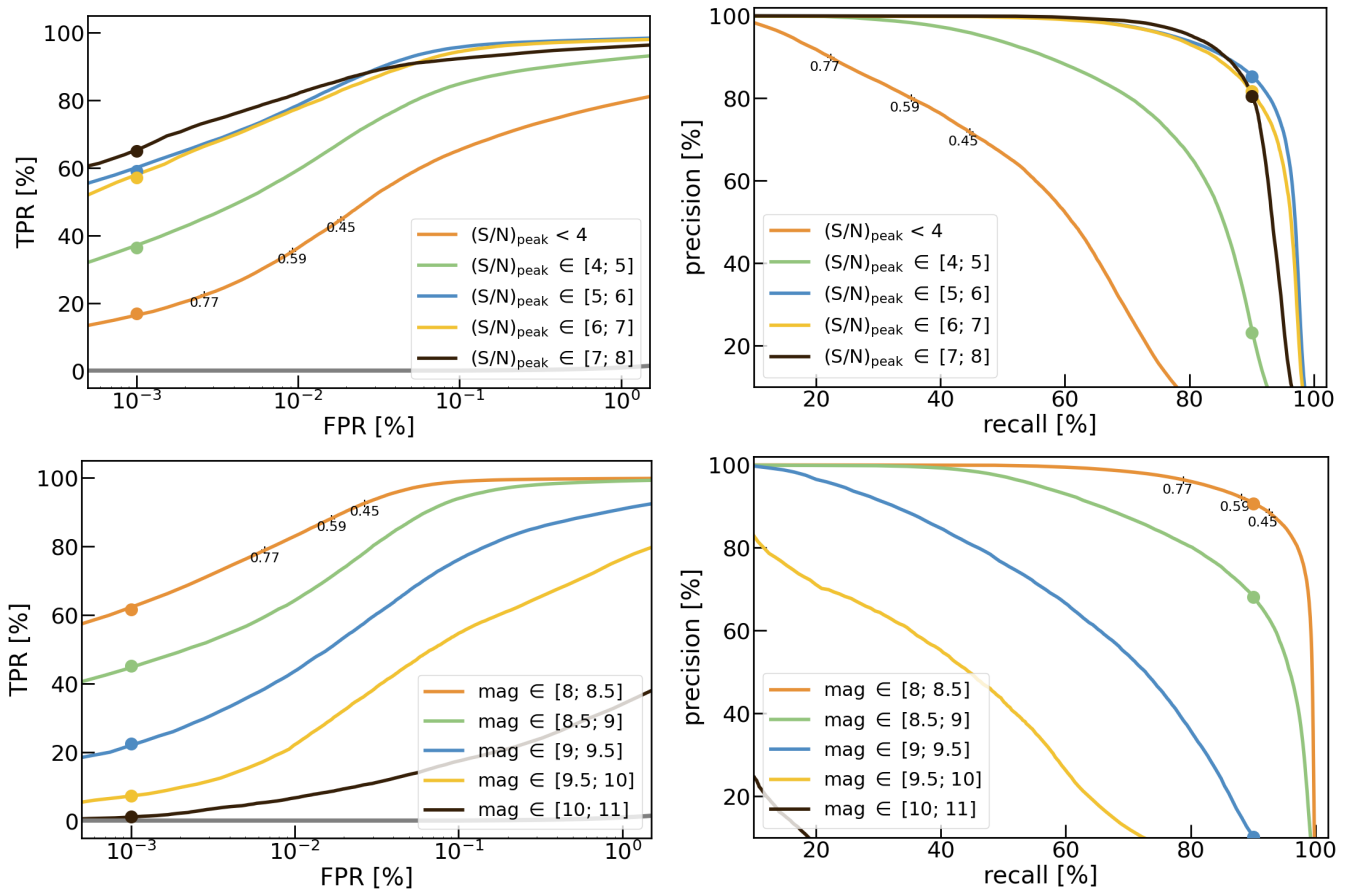


Fig. 7. Performance of UNet_{low} evaluated on set TS_{sim,low}, as a function of streak properties divided into $(S/N)_{\text{peak}}$ bins (*top*), and magnitude bins (*bottom*). *Left*: ROC curves with large dots marking the TPR levels at 0.001% FPR ($\text{TPR}_{0.001}$). *Right*: Precision-recall curves with dots showing the precision at 90% recall (P_{90}). The grey curves indicate a random classifier.

Finally, the main observational artefacts present in Unistellar images are tracking errors due to unstable telescope pointing, and bright star saturation. We visually inspected real and synthetic streaks from TS_{obs} and TS_{sim,high} to find examples of such artefacts, and found a few tens examples, which only allowed for qualitative tests. As shown in Appendix D, UNet-predicted masks tend to be highly stable with respect to both types of artefacts, even for distorted, non-linear streaks that are not simulated with orbitalAI. Other common artefacts such as diffraction spikes are absent from the current Unistellar datasets, and their impact on segmentation performance will require further consideration when retraining and applying the UNet on deeper imaging surveys.

4.4.2. Low flux regime

In this section, we present the performance of UNet_{low}, dedicated to the reconstruction of the faintest satellite streaks with $(S/N)_{\text{peak}} < 8$. To begin, UNet_{low} was applied to simulations from set TS_{sim,high} with $(S/N)_{\text{peak}} < 5$, in order to directly compare both networks. The dotted red curves in Fig. C.1 show significant improvement in the precision of satellite masks from UNet_{low}, with P_{90} increasing to 40.7% with this UNet. $\text{TPR}_{0.001}$ remains comparable between both networks for $(S/N)_{\text{peak}} < 5$, but UNet_{low} gives higher AUC in that bin (0.9813 vs 0.9344). At highest magnitude (mag $\in [9; 10]$), UNet_{low} also improves the

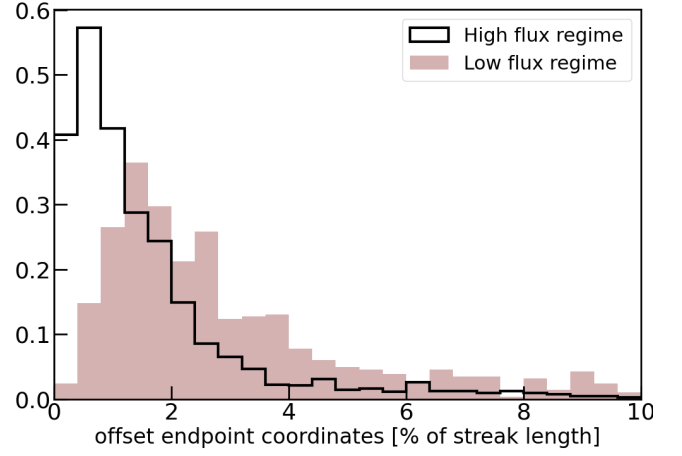
AUROC (0.9603 vs 0.8313), $\text{TPR}_{0.001}$ (17.3% vs 9.7%), and P_{90} (8.0% vs 1.5%). These results demonstrate the benefits of training a separate network for reconstruction of faint streaks.

Similarly to Sect. 4.4.1, we then applied UNet_{low} on set TS_{sim,low}, and computed metrics after separating telescope versions and streaks into $(S/N)_{\text{peak}}$ and magnitude bins. Results are presented in Table 4 and Fig. 7. All metrics gradually increase as a function of S/N, ranging from an AUC of 0.9236, $\text{TPR}_{0.001}$ of 16.9%, and P_{90} of 5.2% at $(S/N)_{\text{peak}} < 4$, to an AUC of 0.993, $\text{TPR}_{0.001}$ of 59.1%, and P_{90} of 85.3% at $(S/N)_{\text{peak}} \in [5; 6]$. At $(S/N)_{\text{peak}} > 6$, we obtain a 3–4% decrease in P_{90} and significantly lower AUC than for lower S/N. This saturation is likely due to the much lower fraction of ground-truth streaks within this bin, as shown with the distribution from Fig. 2.

We emphasize that, in this section, the neural networks are aimed to tackle the challenging task of reconstructing extremely faint satellite streaks, which are mostly invisible by eye. In our simulation pipeline, the width of ground-truth satellite masks is reduced for streaks with lower S/N, but kept constant across the streak elongation. The most common streaks in the training and test sets, with $(S/N)_{\text{peak}} < 5$ (see Fig. 2), are therefore typically near the background noise level, and only a few pixels enclosed within their ground-truth satellite masks are actually detected (see Fig. C.2). As examples, all streaks with $(S/N)_{\text{peak}} < 4$ have an average S/N per pixel within the satellite masks, $(S/N)_{\text{avg}}$, below 0.5. Streaks with $(S/N)_{\text{peak}} \in [4; 5]$ have an $(S/N)_{\text{avg}}$ distribution peaking at 0.3–0.6, and most streaks in the three other bins

Table 4. Evaluation of the UNet performance for segmentation of low-S/N satellite streaks in the test set.

Test set	area under ROC	TPR at 0.001% FPR	precision at 90% recall
all	0.9771	46.2%	40.9%
eVscope 1	0.9842	62.6%	79.2%
eVscope 2	0.9570	45.4%	7.4%
Odyssey	0.9859	29.6%	40.3%
$(S/N)_{\text{peak}} < 4$	0.9236	16.9%	5.2%
$(S/N)_{\text{peak}} \in [4;5]$	0.9727	36.4%	23.2%
$(S/N)_{\text{peak}} \in [5;6]$	0.9935	59.1%	85.3%
$(S/N)_{\text{peak}} \in [6;7]$	0.9916	57.2%	81.6%
$(S/N)_{\text{peak}} \in [7;8]$	0.9855	65.0%	80.6%
$\text{mag} \in [8;8.5]$	0.9992	61.7%	90.6%
$\text{mag} \in [8.5;9]$	0.9973	45.2%	68.1%
$\text{mag} \in [9;9.5]$	0.9756	22.5%	10.2%
$\text{mag} \in [9.5;10]$	0.9286	7.4%	3.4%
$\text{mag} \in [10;11]$	0.7359	1.2%	1.5%

**Fig. 8.** Histogram of the difference between predicted and ground-truth endpoint coordinates of the satellite streaks, for the high- and low-flux regimes. In both cases, the distributions peak at offsets below 3% of the total streak length. A total of 1% and 26% simulated images from the test sets in the high- and low-flux regimes have outlier predictions, with offsets on the endpoint coordinated exceeding 10% of total length.

also have $(S/N)_{\text{avg}} < 1$. It is therefore not surprising that UNet_{low} faces difficulties in extracting the streaks as continuous features, and metrics listed in Table 4 must be interpreted in this context.

To relate the pixel-level precision of output satellite masks with predictions on the overall streak elongations, we followed the procedure from Sect. 4.3 and plotted the difference between predicted and ground-truth endpoint coordinates in Fig. 8. The resulting distribution peaks for differences below 3% of the total streak length, which is only slightly higher than for UNet_{high} evaluated on TS_{sim,high} (black curve in Fig. 8). This shows that, despite predicting truncated satellite masks with several false negative pixels, UNet_{low} recovers the streak endpoint coordinates with good accuracy down to very low S/N. We obtain a total of 26% outliers, where positive pixels predicted by the UNet are associated with artefacts or stars, distant from the actual streak. For typical observing sequences, these outliers can be easily flagged by comparing the position of predicted masks in successive short-exposure images with the expected satellite motions.

To further improve the reconstruction of faint streaks with UNet_{low}, we implemented non-machine learning algorithms to pre-process patches and improve the streak S/N prior to training. We tested a total of four denoising and Gaussian filtering processes, with the baseline UNet architecture, by processing patches in the training, validation sets, and TS_{sim,low}, in order to conduct a direct comparison with the performance of UNet_{low}. Firstly, we implemented total variation denoising (UNet_{denoiTV}), which aims at minimizing the image gradient magnitude with the Chambolle algorithm (Chambolle 2004), adopting a weight parameter of 50. Secondly, we applied Bilateral filtering (UNet_{denoiBilat}) which conducts a weighted average of nearby pixels, as measured by the Gaussian function of the relative pixel separation, with a standard deviation of 1 pix. Thirdly, we denoised patches based on the non-local means algorithm (Buades et al. 2005) that approximates a given central pixel with values from neighboring pixels (UNet_{denoiNLM}). The algorithm uses a similarity criteria between windows centered on the central pixel, and windows centered on the surrounding pixels. We adopted a window size of 3 pixels and a maximum distance of 6 pixels to select windows for denoising. In these three cases, denoising parameters were chosen as a tradeoff between optimisation of output $(S/N)_{\text{peak}}$ distributions, and introduction

of image artefacts. The last method for reducing high-frequency noise components is a Gaussian filtering with a kernel of standard deviation 1.5 pix (UNet_{GaussSmooth}).

Sequential denoising and segmentation shows significant improvement for the segmentation of images in set TS_{sim,low}, in particular with UNet_{denoiTV} and UNet_{denoiNLM}. Figure 9 illustrates the significant increase in metrics obtained with these two networks, for the same $(S/N)_{\text{peak}}$ bins as in Fig. 7. The AUC after denoising are >0.99 in every bin down to $(S/N)_{\text{peak}} < 4$, and Fig. 9 shows a drastic increase in P_{90} for the predicted satellite masks, compared to the baseline UNet_{low}. For instance, at $(S/N)_{\text{peak}} \in [4;5]$, the precision increases from 23.2% with UNet_{low} to 70.3% and 62.2% with UNet_{denoiTV} and UNet_{denoiNLM}, respectively. Both methods also improve the precision by $\approx 15\%$ in the $(S/N)_{\text{peak}} \in [7;8]$ bin. In fact, denoising mainly helps recover higher numbers of positive pixels enclosed within the satellite masks, down to very low S/N. This results in fewer discontinuities and improved mask morphologies (see Appendix C). The $\text{TPR}_{0.001}$ generally, but not systematically, increases by a few percent.

We caution that this gain in performance requires to flag and reject images with significant artefacts introduced by the algorithms, by imposing a higher streak $(S/N)_{\text{avg}}$ after denoising. Novel deep learning algorithms have been developed to conduct joint denoising and segmentation (e.g., DenoSeg, Buchholz et al. 2020), based on a self-supervised denoising scheme. The implementation of such methods will be particularly interesting to reduce the dependence on artefacts.

5. Application to deeper single-visit images

After testing on shallow, short-exposure images from Unistellar telescopes, in this section, we probe the generalisation capabilities of the overall pipeline to detect and reconstruct satellites streaks in deeper images. We evaluate the UNet_{high} model on single-visit data products from the HSC Transient Survey (Yasuda et al. 2019) conducted with the Subaru 8-m telescope, in order to quantify its ability to flag streak pixels, for images differing drastically from those seen during training.

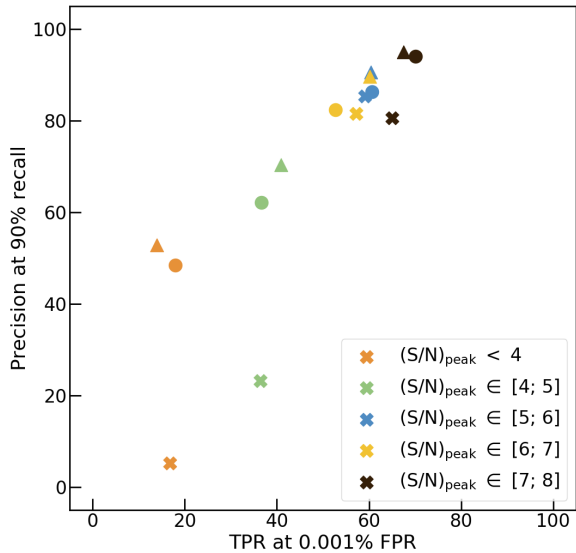


Fig. 9. Scatter plot showing the evolution of the two main metrics, the precision at 90% recall (P_{90}), and the TPR at 0.001% FPR ($TPR_{0.001}$), for the baseline $UNet_{low}$ (crosses), and for networks trained and tested on images denoised with non-ML algorithms. Triangles and circles correspond to denoising with the total variation approach ($UNet_{denoisTV}$), and with non-local means algorithm ($UNet_{denoisNLM}$), respectively. Performance are shown for the same $(S/N)_{peak}$ bins as in Fig. 7.

The HSC Transient Survey is a cadenced survey of the COSMOS field reaching depths of 26.4, 26.3, 26.0, and 25.6 mag in g -, r -, i -, and z -bands, respectively, and with pixel sizes of $0.168''$. Instead of HSC coadded data products, we used single-visit images of 300 s with depths of ≈ 22 mag and focused on i -band images. This provides an ideal data set for testing the UNet on images with much higher densities of astronomical sources, and different artefacts and background properties.

We started from the list of HSC i -band exposures with detected satellite streaks that were flagged and kindly shared by C. Saunders. The PSF-matched, "warped" images of regions covering the streaks were imported with the Data Archive System. We then produced difference images by subtracting the next PSF-matched warp from each warp containing a streak. This resulted in a data set of 33 difference images with satellite streaks. We then applied $UNet_{high}$ to these images using overlapping sliding windows (see Sect. 3.3.2).

In these cases, ground-truth satellite masks are available by comparing a given exposure with other exposures of the same field. Streak pixels are flagged by the `CompareWarpAssembleCoadd` algorithm from the HSC pipeline through this type of temporal filtering. When applied to difference images, $UNet_{high}$ recovers $>90\%$ pixels enclosed within the satellite masks, without any false positive pixel across the images. This shows the promising performance of our pipeline on much deeper images than those used in the training and validation sets. In addition, Fig. 10 illustrates the network robustness for excluding residual sources after image differencing, regardless of their morphology. Reconstructed streaks are also comparable to masks produced by the traditional algorithms implemented in the HSC pipeline.

This suggests that the UNet model includes a generic representation of linear streaks which, in addition to the arcsinh normalisation of patches, and to the sliding window approach, make

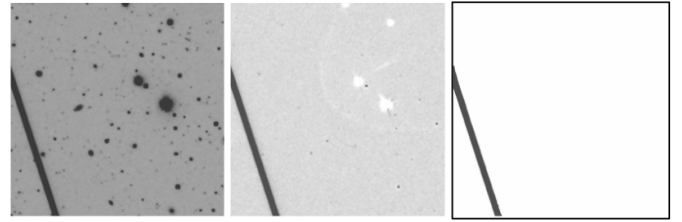


Fig. 10. Demonstration of the generalization capabilities of $UNet_{high}$, when applied to Subaru/HSC i -band exposures of ≈ 22 mag depth, after PSF matching and image differencing. We show the original 300 s exposure with a detected streak (left), the difference image (middle), and the satellite mask predicted by $UNet_{high}$ (right). Despite training on very different image depth and quality, the UNet recovers the correct elongation of the streak, and does not provide any false positive pixel at the position of image difference residuals.

it useful for a broad range of astronomical data sets. Overall, this suggests that the streak simulation package and the deep-learning pipeline presented in this paper are versatile enough for a future retraining and optimization for deep, wide-scale surveys.

6. Conclusions

In this work, we developed a deep learning pipeline in the PyTorch framework to identify satellite streaks in wide-field, single-band images, and help reduce the impact of satellite constellations on astronomical observations. **To reduce the inference time on large, unbalanced data sets**, we followed a two-step approach including (1) traditional convolutional neural networks (CNN) to rapidly classify images with streak candidates, and (2) fully-convolutional networks (UNet) to conduct the pixel-level reconstruction of satellite streaks. Realistic simulations of satellite streaks were obtained with the `orbitAI` Python package to train the neural networks. **We aimed at quantifying and stabilizing the performance as a function of streak properties, and focused on optimizing the mask reconstructions down to the lowest signal-to-noise ratios, in order to complement existing streak detection algorithms (e.g., Paillassa et al. 2020; Stoppa et al. 2024; Liu et al. 2025).**

We focused on targeted observing campaigns of LEO satellites with the network of Unistellar telescopes, in order to train and test the pipeline under broad ranges of observing conditions, while also including three telescope versions with different pixel scales. For the CNN classifier, we measure a TPR of nearly 90% on real streaks detected by amateur astronomers, while keeping the FPR below 1%. Most of the false negatives from the CNN are truncated streaks crossing image borders. Further testing on simulations show that the selection of satellite streaks with Unistellar sequences of short-exposures images is nearly complete up to $\text{mag} \approx 8$. With this low false-negative rate, the CNN efficiently flags interesting images for segmentation with the UNet.

With the baseline UNet, $UNet_{high}$, optimized for streaks with a peak S/N above 5, we recover $>80\%$ pixels within the ground-truth satellites masks, for a FPR per pixel of 0.001% (i.e. maximum 10 FP pixels per image). For 90% of streak pixels correctly recovered, the precision of output satellite masks is $>96\%$ at $(S/N)_{peak} \in [5; 8]$, and nearly 100% at $(S/N)_{peak} \geq 8$. Moreover, $UNet_{high}$ shows nearly-ideal predictions of the streak elongations, as required to reconstruct the actual LEO satellite motions for space surveillance. We obtain differences between predicted and ground-truth endpoint coordinates peaking below 2% of the

total streak length, and only $\leq 1\%$ catastrophic outliers with differences $> 10\%$. We trained a separate UNet, UNet_{low}, to optimize the reconstruction of the faintest streaks, finding that the precision at 90% recall increases to 40.7% at $(S/N)_{\text{peak}} < 5$.

Both neural networks are implemented in the Unistellar data analysis pipeline and currently conducting the detection, segmentation, and photometry of streaks over 30 to 60 new sequences of satellite observations per week (i.e. $> 10\text{k}$ images/week). **With a total inference time of 10 min per sequence, our pipeline enable efficient analyses of the data collected by Unistellar telescope users, which helps monitoring satellite positions and brightnesses, and will eventually help populate public databases of LEO satellite observations.** Moreover, we tested the UNet on much deeper single-visit images from the Subaru 8-m telescope without any retraining, obtaining good performance in comparison to the original non-ML algorithm based on the Hough Transforms. Given these generalization abilities, we expect that the pipeline will reach nominal performance with retraining, as requested for future applications to deep, wide-scale surveys such as *Rubin* LSST.

In the future, complementary deep-learning approaches could enhance the performance of satellite streak detection and masking in short-exposure images. When larger numbers of sequences become available for training, one can start conducting the detection based on the entire image time-series with convolutional long-short term memory networks (ConvLSTM; e.g., Shi et al. 2015). Moreover, performing the direct segmentation of image time-series with temporal convolutional networks (TCN; e.g., Lea et al. 2016) or with VNet-like architectures (Milletari et al. 2016) could also help account for the spatio-temporal information encoded in the streak motion.

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Appendix A: Neural network architectures

This section presents the neural network architectures of the baseline CNN for image classification (Fig. A.1), and the baseline UNet for the pixel-level segmentation of satellite streaks (Fig. A.2).

Appendix B: Performance of the streak reconstructions

This section contains more details on the performance of the UNet in the high- and low-flux regimes, with ROC and precision-recall curves separated into different S/N bins and into individual telescopes.

In Fig. B.1, we characterize the influence of the telescope version on the performance of UNet_{high}, for various $(S/N)_{\text{peak}}$ bins. Variations in metrics are relatively small when averaging over the entire test set, but remain more prominent for the faintest simulated streaks in TS_{sim,high}. For the faintest bin with $(S/N)_{\text{peak}} < 5$, TPR_{0.001} is 49.8%, 42.2%, and 29.2% for eVs-scope 1, eVs-scope 2, and Odyssey images, respectively, and P₉₀ is 49.5%, 15.2%, and 10.1%. For the $(S/N)_{\text{peak}} \in [5;8]$ bin, metrics get more stable, with a TPR_{0.001} value of 83.0%, 81.6%, and 79.6% for eVs-scope 1, eVs-scope 2, and Odyssey images, respectively, and P₉₀ of 98.7%, 98.2%, and 94.3%.

In Fig. B.2, we further test UNet_{low} as a function of telescope version, choice of the UNet architecture, and application of image denoising prior to segmentation.

Firstly, metrics appear to strongly depend on the telescope type. TPR_{0.001} and P₉₀ increase by 33% and 38.9%, respectively, between test Odyssey and eVs-scope 1 images. This suggests a stronger dependence than measured for UNet_{high} (see Fig. B.1), probably due to significant differences in the (S/N) distributions of the parent training, validation, and test samples (see Fig. 2). Interestingly, P₉₀ drops to $\leq 10\%$ for eVs-scope 2.

Secondly, when averaging over the telescope version, we obtain lower performance for the two alternative architectures with respect to the baseline. The AUC, TPR_{0.001}, recall at 90% recall are 0.9731, 38.6%, 17.0% for UNet_{noskip}, and 0.9733, 38.9%, 18.8% for UNet_{++mod}, respectively, compared to 0.9771, 46.2%, and 40.9% for the baseline UNet adopted in our analysis.

Thirdly, using sequential denoising plus segmentation with the baseline UNet architecture shows some improvement in the overall streak reconstruction accuracy. The value of TPR_{0.001} is 46.2%, 52.5%, 51.3%, 47.5%, and 52.3%, and the value of P₉₀ is 40.9%, 84.2%, 78.1%, 76.5%, and 81.2% for UNet_{low}, UNet_{denoiTV}, UNet_{denoiNLM}, UNet_{denoiBilat}, and UNet_{GaussSmooth}, respectively.

The difference between the predicted and ground-truth endpoint coordinates of the satellite streaks is plotted in Fig. B.3, to illustrate its evolution as a function of streak S/N. For the four bins with $(S/N)_{\text{peak}} > 5$, the distributions for UNet_{high} peak at offsets below 2% of the total streak length, and most simulations remain below 3%. At $(S/N)_{\text{peak}} > 5$, there is a longer tail of streaks extending to 10%. For UNet_{high}, the fraction of outliers per bin are 37.7%, 10.0%, 2.4%, 1.2%, and 0.9%, respectively, from the lowest- to highest- (S/N) . For UNet_{low}, we also find distributions peaking for small offsets of only 1–3% of the total streak length. The distributions are broader than for UNet_{high}, especially for the bin with $(S/N)_{\text{peak}} < 5$ which remains roughly flat. For this UNet, the fraction of masks with offsets on the endpoint coordinated exceeding 10% of total length are 50.0%, 30.9%, 21.2%, 7.3%, and 8.5%, respectively, from the lowest- to highest- (S/N) bins. This illustrates the limitations of UNet_{low} in predicting accurate

satellite masks for extremely tenuous streaks, when no denoising pre-process is implemented to boost the segmentation task.

Appendix C: Examples of satellite masks

Examples of output satellite masks produced by UNet_{high} and UNet_{low} are shown in Fig. C.1 and C.2, for satellite streaks in different S/N regimes. In particular, Fig. C.2 illustrates the challenging task of UNet_{low}, which is trained to reconstruct satellite masks for very faint streaks typically near the background noise level. In the low-S/N regime, conducting image denoising prior to segmentation helps recover higher numbers of positive pixels within the satellite masks (see Fig. C.3).

Appendix D: Artefact handling

In this section, we illustrate the performance of the UNet in presence of typical observational artefacts. The list of common artefacts and their analysis result from a visual inspection of real and synthetic streaks in sets TS_{obs} and TS_{sim,high}. Firstly, we noticed that a subset of Unistellar observing sequences present pointing instabilities, with tracking errors producing extended (streak-like) stellar light profiles and/or distorted (non-linear) streak morphologies. These instabilities are typically present in $< 5\%$ of sequences, and affect at most 10% images from these sequences. In the rare cases where pointing instabilities affect a detected streak from TS_{obs}, UNet_{high} is able to correctly segment the distorted streak and to exclude the elongated stars from the output satellite mask (see Fig. D.1). Secondly, bright saturated stars are present in some fields-of-view and we inspected the ability of UNet_{high} to classify the few saturated pixels near the star centroids. Although a few pixels have elevated UNet scores, they are eventually excluded during binarization, given the requirement of > 64 contiguous pixels per positive region (see Fig. D.1).

Given the significant number of streaks overlapping stars in our training sets, the output satellite masks are generally robust with respect to blending. In Fig. D.2, we show two successful examples of streaks from TS_{sim,high} with stars overlapping the injected satellite streaks over a few pixels. In both cases, the streak and blended stars have comparable peak (S/N) , which helps the UNet deblend both components. When the stellar over streak luminosity ratio increases, the UNet has more difficulties to produce accurate masks and tends to either include or mask out both components. This is a typical failure case of UNet_{high}.

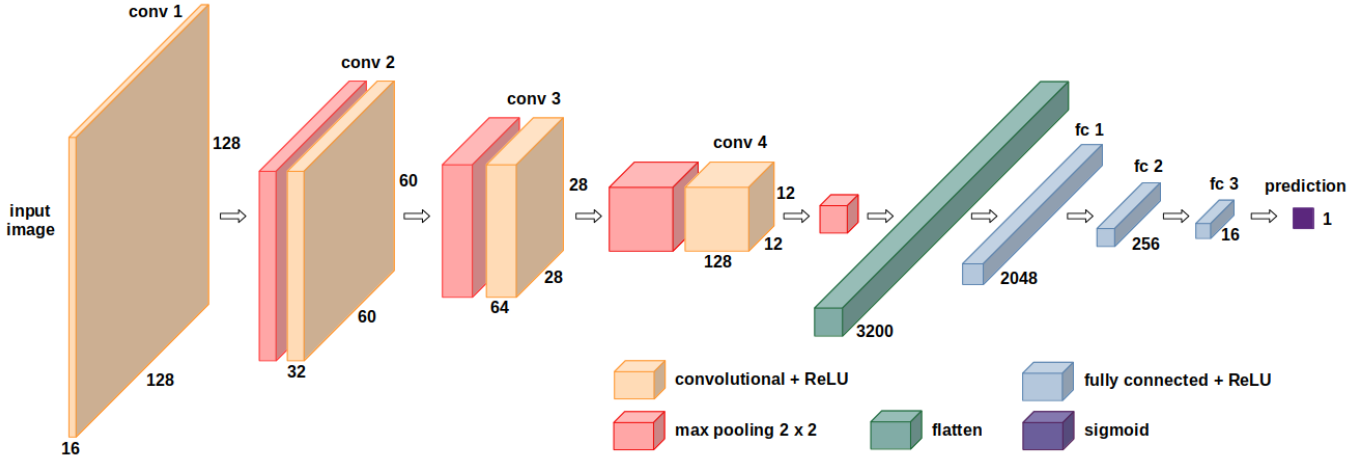


Fig. A.1. Architecture of the baseline convolutional neural network for classification of satellite streaks. The four convolutional layers have kernel sizes of 9×9 , 5×5 , 5×5 , and 3×3 , respectively, with 16, 32, 64, and 128 filters. Three fully-connected (FC) hidden layers have 2048, 256, and 16 neurons, respectively. Max-pooling layers with 2×2 kernels and stride = 2 are used between convolutional layers, and ReLU activations are applied between each layer.

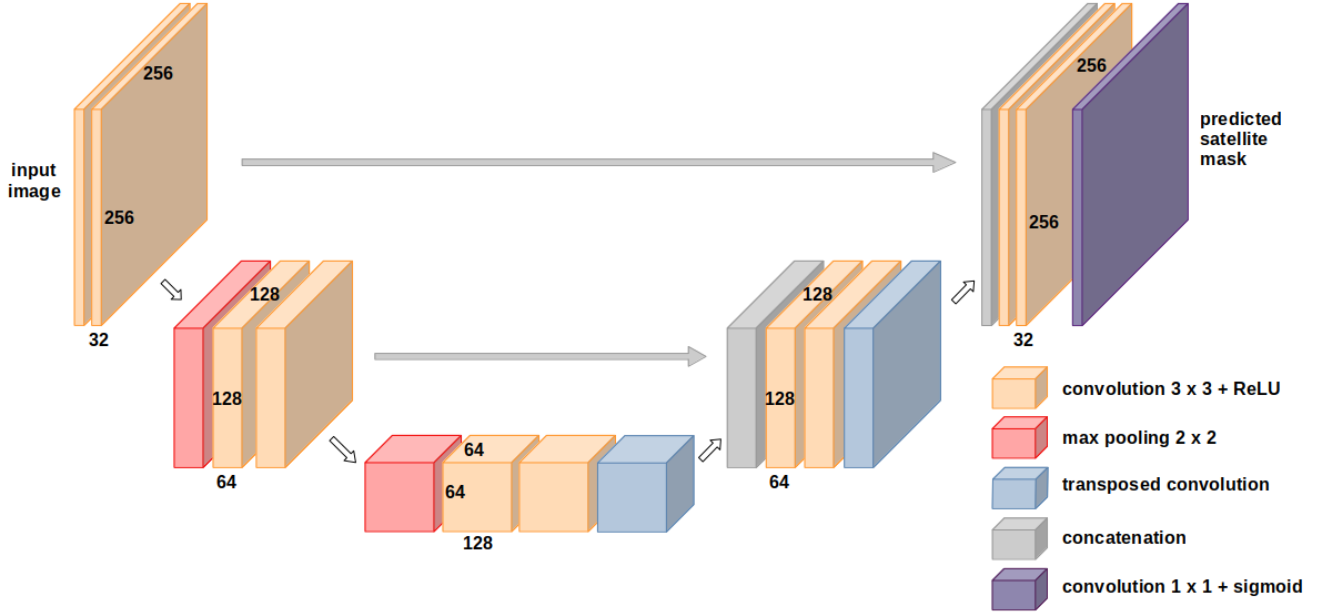


Fig. A.2. Architecture of the baseline UNet for segmentation of satellite streaks. The encoder comprises two blocks with two 3×3 convolutional layers with ReLU activations, followed by a 2×2 max-pooling layer. The decoder includes two blocks with a transposed convolution with stride of 2, followed by two 3×3 convolutional layers with ReLU activations. For input patches of 256×256 pixels, we obtain dimensions of $64 \times 64 \times 128$ after the bottleneck. Skip connections between the encoder and decoder are represented with grey arrows. The output layer is a 1×1 convolution layer with sigmoid activation.

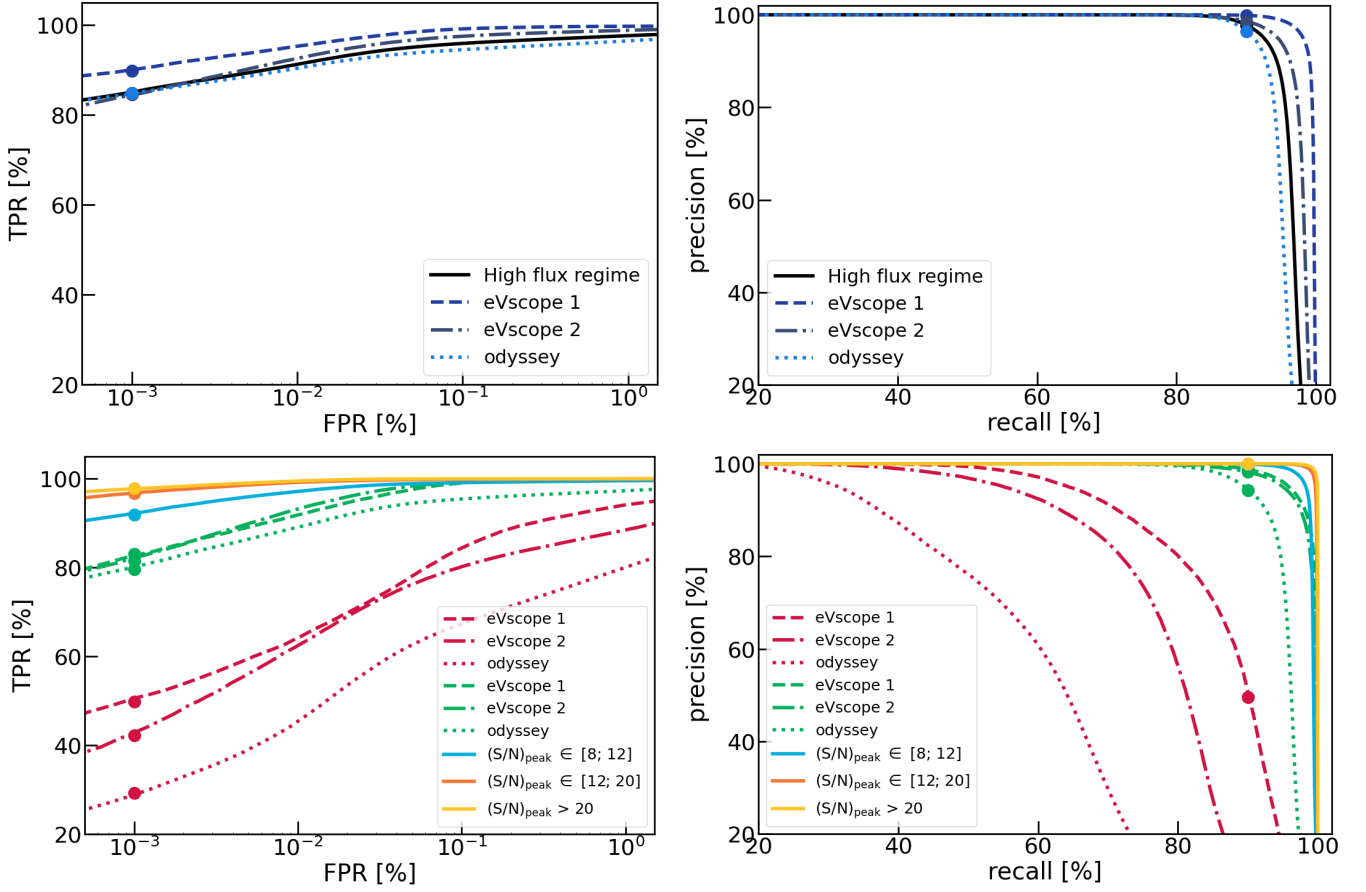


Fig. B.1. Performance of UNet_{high} evaluated on set TS_{sim,high}, as a function of telescope version. *Left:* ROC curves with large dots showing the TPR levels at 0.001% FPR (TPR_{0.001}). *Right:* Precision-recall curves with dots marking the precision at 90% recall (P₉₀). *Top:* Performances averaged over all eVscope 1, eVscope 2, and Odyssey simulations from TS_{sim,high}. *Bottom:* Variations in the two bins with lowest (S/N)_{peak} from Fig. 6, namely (S/N)_{peak} < 5 (red curves), and (S/N)_{peak} ∈ [5;8] (green curves), for eVscope 1 (dashed curves), eVscope 2 (dash-dotted curves), and Odyssey (dotted curves) images. For clarity, for the last three bins, we only show the performance averaged over the entire test set (plain curves).

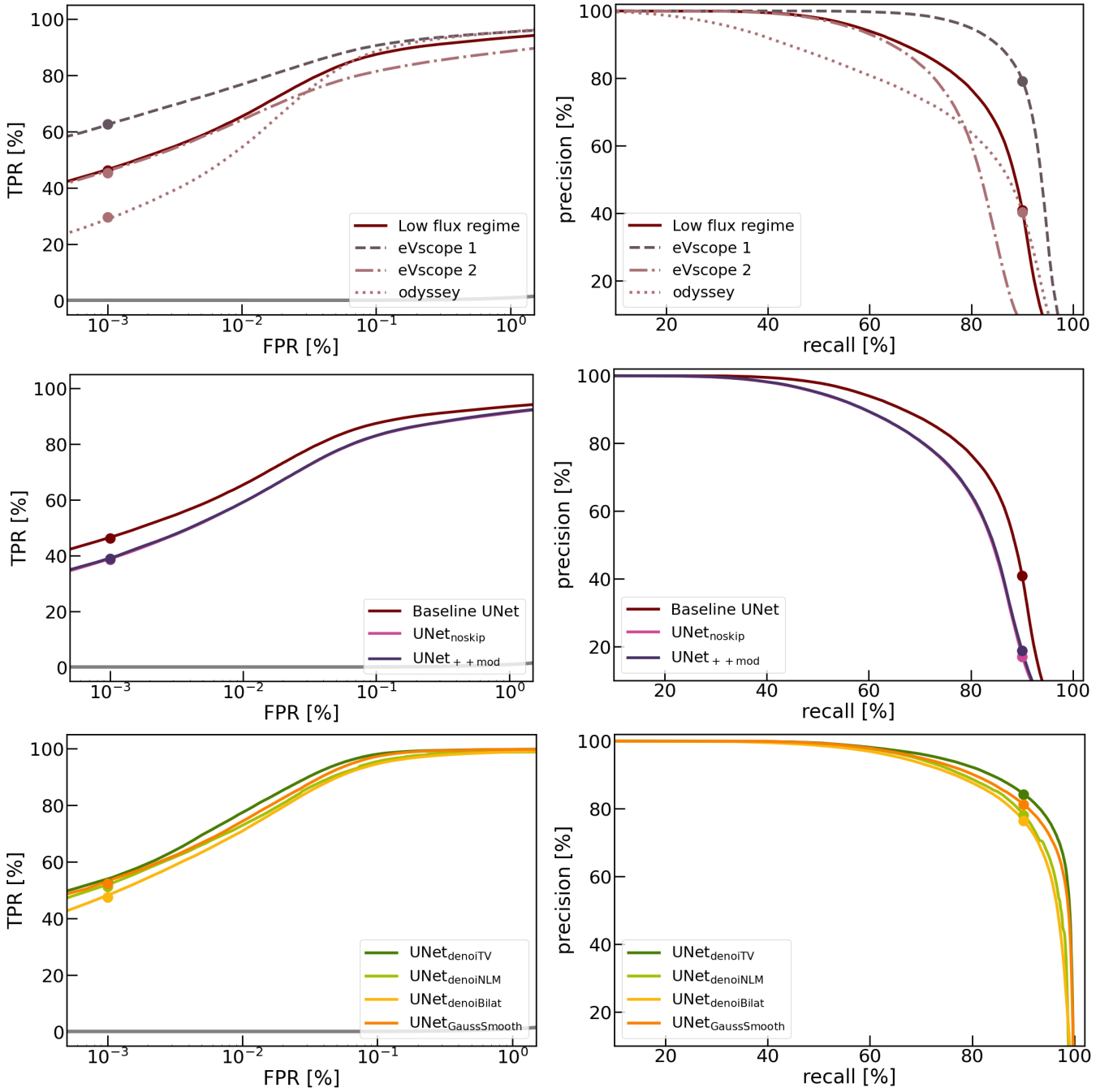


Fig. B.2. Performance of UNet_{low} evaluated on set $\text{TS}_{\text{sim,low}}$, with the baseline UNet shown in all panels for reference (purple curves). *Left:* ROC curves with large dots showing the TPR levels at 0.001% FPR ($\text{TPR}_{0.001}$). The grey curves indicate a random classifier. *Right:* Precision-recall curves with dots marking the precision at 90% recall (P_{90}). *Top:* Evolution for eVscope 1, eVscope 2, and Odyssey simulations from $\text{TS}_{\text{sim,low}}$. *Middle:* ROC and precision-recall curves for the baseline UNet and two alternative architectures, $\text{UNet}_{\text{noskip}}$ and $\text{UNet}_{++\text{mod}}$ (see details in Sect. 3.3), when combining all three telescopes. *Bottom:* Implementation of different denoising algorithms as pre-processing (see text for details).

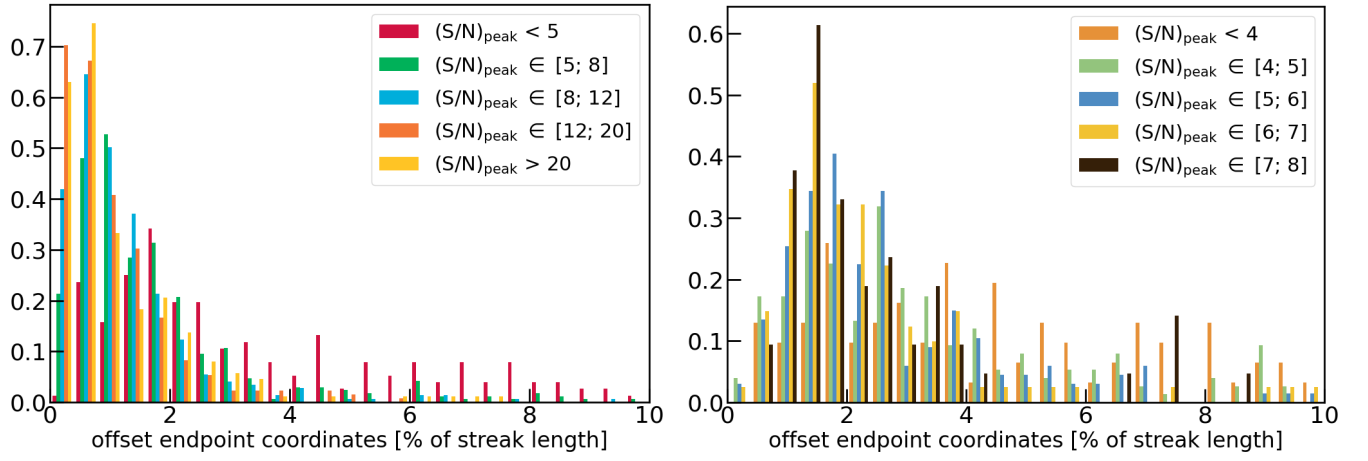


Fig. B.3. Difference between the predicted and ground-truth endpoint coordinates in various bins for UNet_{high} (left) and UNet_{low} (right).

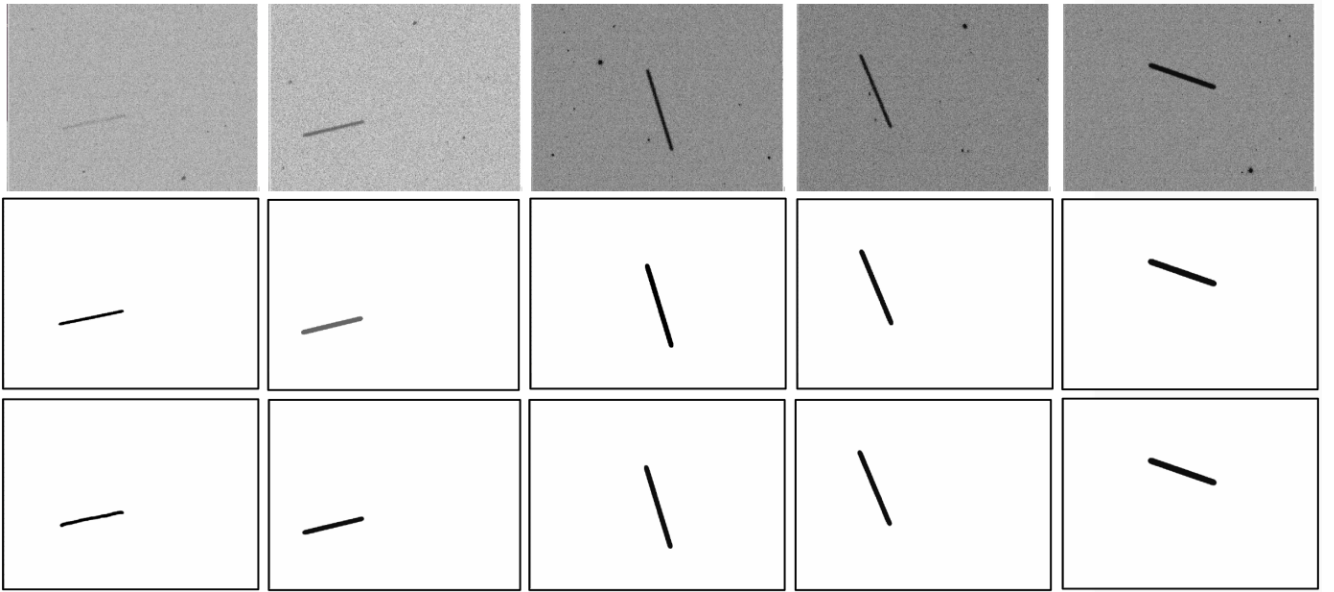


Fig. C.1. Examples of segmentation of satellite streaks in the high-flux regime with UNet_{high}. The top, middle and bottom rows show the simulated images with injected streak, the ground-truth satellite mask, and the mask predicted by UNet_{high}, respectively. Each column shows an example from the $(S/N)_{\text{peak}}$ bins defined in Fig. 2. The five streaks have $(S/N)_{\text{peak}}$ of $\approx 4, 7, 9, 15$, and 50 , respectively.

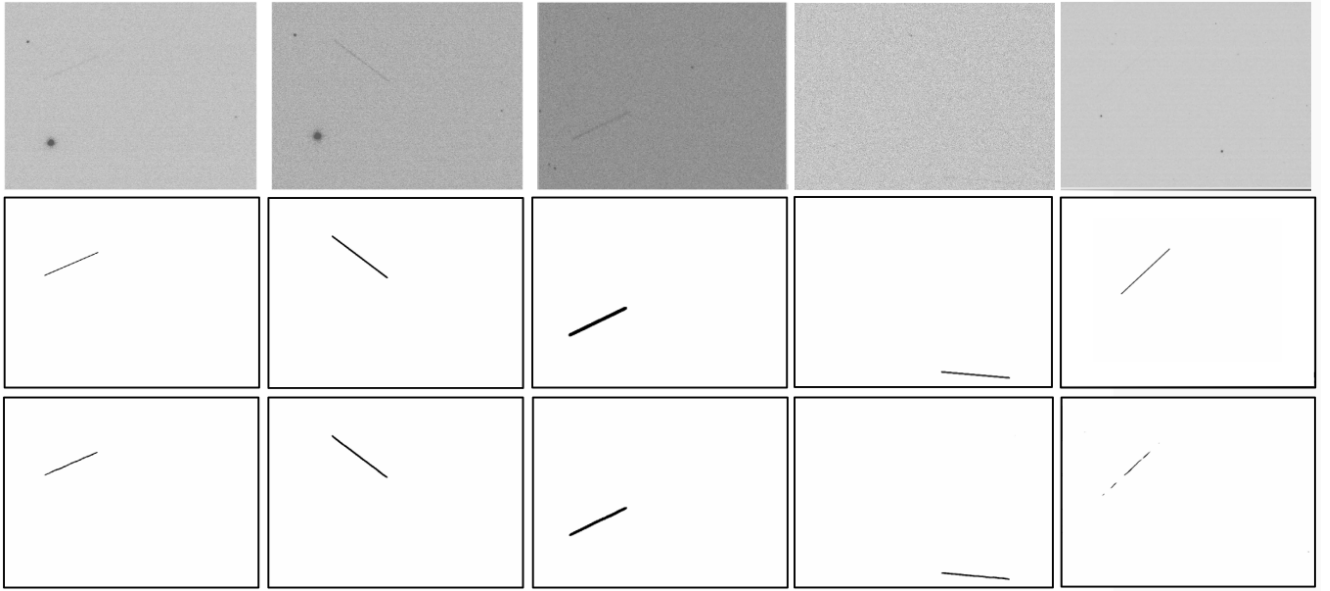


Fig. C.2. Examples of segmentation of satellite streaks in the low-flux regime with UNet_{low} . The top, middle and bottom rows show the simulated images with injected streak, the ground-truth satellite mask, and the mask predicted by UNet_{low} , respectively. The five streaks shown as examples have $\text{SNR}_{\text{peak}} \lesssim 5$. The two rightmost panels illustrate the good performance of UNet_{low} to reconstruct the positions, elongations, and pixel-level masks for the faintest streaks invisible by eye in the *orbitAI* simulations, which only have a few pixels from the ground-truth satellite masks with $(\text{S/N}) > 3$.

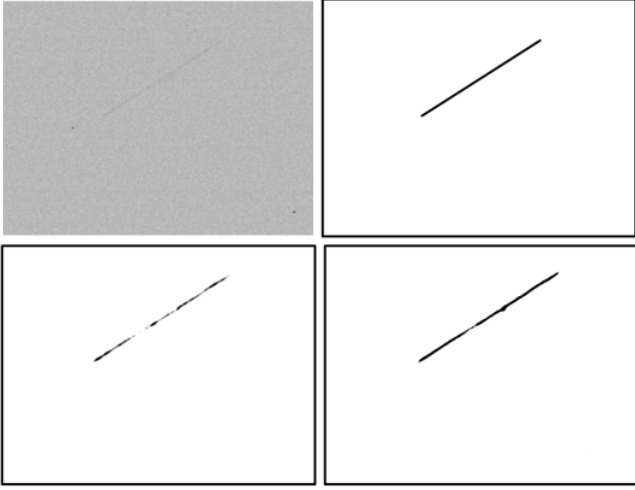


Fig. C.3. Illustration of the improvement obtained with image denoising for a streak with $(S/N)_{\text{peak}} \simeq 4$ and $(S/N)_{\text{avg}} \simeq 0.3$. The top subpanels show the simulation and ground-truth mask from `orbitAI` and the bottom left and right panels show the streak reconstructions with the baseline `UNetlow` and `UNetdenoisTV`, respectively.

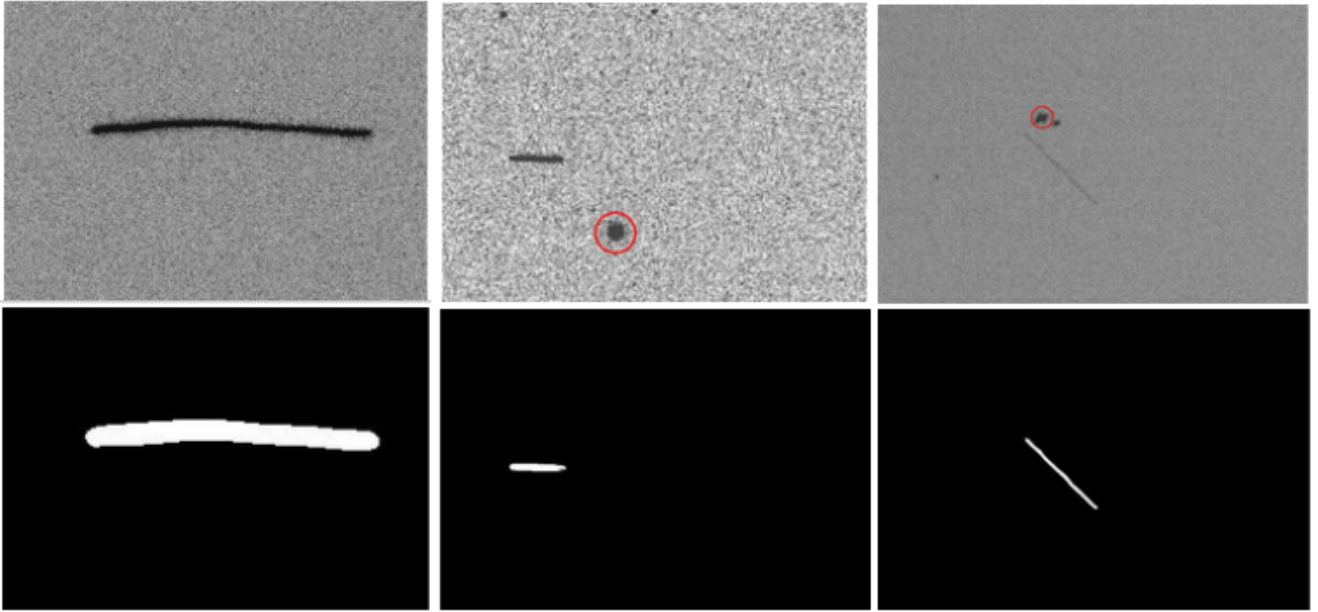


Fig. D.1. Stability in presence of artefacts. Illustration of the segmentation of a real STARLINK streak from TS_{obs} , with our high-flux UNet, in the presence of pointing instabilities (*left*). Examples of segmentation of satellite streaks with nearby saturated stars (red circles), for a real STARLINK streak from TS_{obs} (*middle*) and a synthetic streak from $TS_{\text{sim,high}}$ (*right*). The top and bottom rows show the simulated images and the masks predicted by UNet_{high}, respectively.

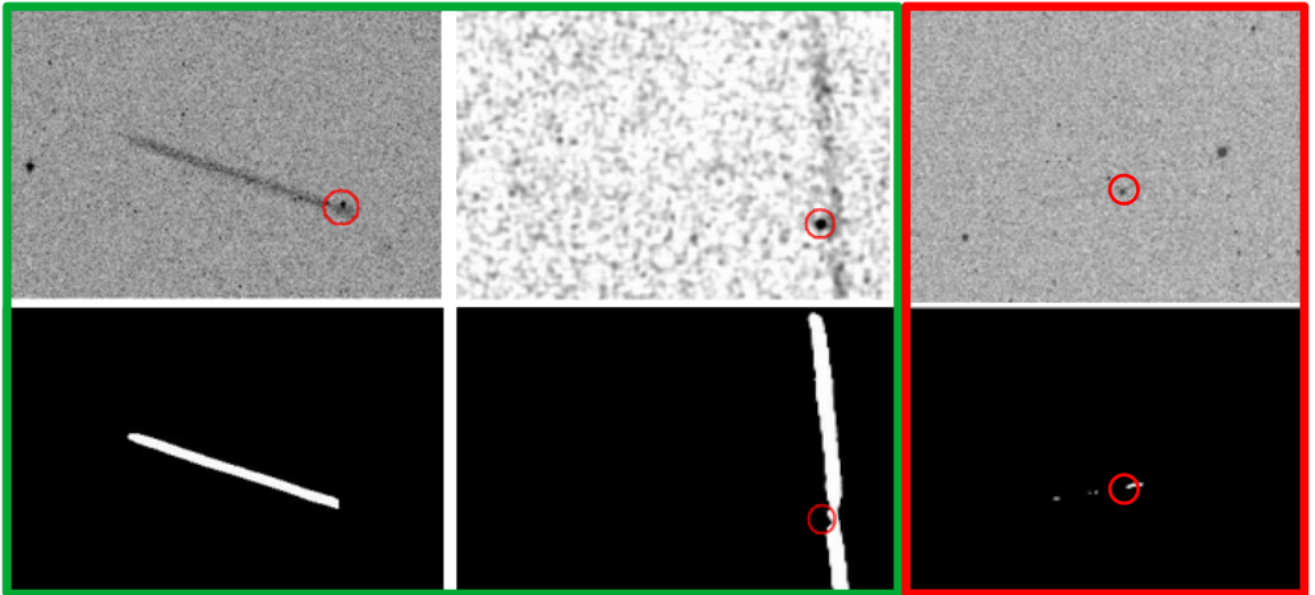


Fig. D.2. *Left and middle:* Two examples of blended streaks, where nearby stars (red circles) are correctly excluded from the output satellite masks. *Right:* Illustration of a typical failure case comprising a low-(S/N) streak blended with a bright star. All panels show zoom-in views of the entire images in order to focus on the region of interest.